

Establishing a Unified Optimal Investment Strategy for Defined Contribution Pension Plans:

Clipped versus Directly Constrained Mean–Variance Controls and a Three-Layer Glide-Path Design

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Abstract

Defined contribution (DC) pension investment is often discussed by comparing exogenously specified glide paths, including target-date, reverse, and U-shaped designs. Such comparisons do not establish which strategy is optimal once member objectives, time inconsistency, contribution cash flows, and practical investment constraints are considered. This paper develops a unified framework in which glide paths arise endogenously from constrained dynamic optimisation.

We examine four mean–variance solution concepts: pre-commitment mean–variance (PCMV), dynamically optimal mean–variance (DOMV), time-consistent mean–variance with constant variance aversion (cTCMV), and time-consistent mean–variance with total-pension-wealth-dependent variance aversion (dTCMV). Risky-asset investment is constrained between zero and the current DC account balance, ruling out short selling and borrowing against future contributions.

For each concept, we distinguish an unconstrained benchmark, a clipped approximation obtained by projection onto the feasible range, and the directly constrained feedback obtained by incorporating the constraint into the continuation-value problem. Verification theorems, free-boundary characterisations, and a unified Markov–Gauss–Hermite backward–forward engine provide a common theoretical and numerical basis for comparison.

We also introduce a rolling conditional outcome profile. The law of total variance decomposes time-zero terminal variance into the expected conditional variance remaining after an intermediate date and the cross-state variance of conditional expected terminal wealth. By fixing the observed current balance, the rolling profile separates wealth heterogeneity already created by past outcomes from the prospective effect of the future investment rule. It reports conditional mean, dispersion, quantiles, tail risk, skewness, and current risky allocation, while common-state comparisons isolate the effect of changing strategy from the same member state.

The results show that target-date, U-shaped, and intermediate glide paths can emerge endogenously. Clipping accuracy differs materially across solution concepts. At a matched mean, time-zero comparisons naturally favour PCMV because it minimises terminal variance under commitment, whereas rolling evaluation reveals state-specific trade-offs hidden by aggregate rankings, particularly the additional upside participation and downside risk generated by dTCMV.

For mean–variance–skewness extensions, cTCMVS coincides with cTCMV in the constant-coefficient unconstrained Black–Scholes case, and the corresponding clipped controls also coincide; this equality is not established for directly constrained controls. dTCMV–MVS can materially reshape the timing of risk-taking, while PCMV–MVS and DOMV–MVS remain exploratory because of non-concavity and numerical sensitivity.

Accordingly, we propose a three-layer design: cTCMV as the standard default, DOMV for periodic review, and dTCMV as a governed personalised advisory layer, with PCMV serving as a design benchmark and MVS retained as a research extension.

Keywords: defined contribution pensions; retirement-income security; dynamic mean–variance optimisation; directly constrained feedback; clipped approximation; free boundary; rolling conditional evaluation; law of total variance; unified Markov–Gauss–Hermite engine; mean–variance–skewness; three-layer design.

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1 Introduction

DC pension investment is commonly discussed by comparing exogenously specified glide paths, such as conventional target-date, reverse, and U-shaped strategies. Although such comparisons are useful, they do not identify which strategy is optimal once member objectives, time inconsistency, and implementable investment constraints are taken seriously. This paper develops a unified framework in which glide paths are generated endogenously by constrained dynamic optimisation.

Mean–variance optimisation is adopted as the core objective. Because dynamic mean–variance problems are time-inconsistent, optimality is not uniquely defined. We therefore consider four solution concepts: pre-commitment mean–variance (PCMV), dynamically optimal mean–variance (DOMV), time-consistent mean–variance with constant variance aversion (cTCMV), and time-consistent mean–variance with total-pension-wealth-dependent variance aversion (dTCMV). For each concept, we distinguish the unconstrained solution, the clipped approximation obtained by projecting it onto the feasible range, and the directly constrained feedback derived from the constrained problem itself. The central objective is to identify solution concepts that balance member outcomes and implementability, and to translate the four mathematical concepts into a practical three-layer glide-path design.

1.1 The global expansion of DC pensions and the actuarial responsibility

Retirement-income systems are undergoing a structural transition from defined benefit (DB) plans towards DC and other individual-account arrangements. OECD data for 44 jurisdictions show that the share of occupational pension assets held in DB plans declined from 39.7% in 2014 to 31.9% in 2024. In 2024, occupational DC assets grew by 11.4%, compared with 4.0% for occupational DB assets ([OECD, 2025](#)). In the seven largest pension markets, DC arrangements accounted for 63% of pension assets at the end of 2025, and DC assets reportedly grew at an annual rate of 9.4% over the preceding decade ([Thinking Ahead Institute, 2026](#)). This is more than a change in plan labels or asset volumes. Under DB plans, investment risk, underfunding risk, and much of the uncertainty surrounding retirement income have traditionally been managed collectively by the plan sponsor. Under DC plans, a substantial part of these risks is transferred to individual account balances and decisions. Default investment design, glide paths, and retirement-readiness diagnostics are therefore actuarial issues affecting the retirement security and welfare of large populations, rather than merely matters of personal portfolio management.

Pension actuarial practice historically developed around DB benefit valuation, funding methods, contribution setting, funded status, and sponsor risk management. The same actuarial toolkit includes long-horizon cash-flow projection, stochastic modelling, the interaction of assets with liabilities and income, multi-period risk assessment, assumption setting, sensitivity analysis, model validation, and communication of model limitations. These capabilities are directly relevant to DC plans in which members bear investment risk. The International Actuarial Association (IAA) recognises the long-standing central role of actuaries in DB plan management and notes that, as systems move from DB to DC, skills developed in DB practice are increasingly applied to DC product design, investment strategy, and longevity-risk management ([International Actuarial Association, 2021](#)). It also identifies the development and application of models for complex financial problems, together with professional explanation of their practical implications and limitations, as core actuarial capabilities ([International Actuarial Association, 2013](#)).

The IAA describes the profession’s purpose as enabling the future financial well-being of people, organisations, economies, and society, and emphasises long-term consequences, the public interest, and positive social impact ([International Actuarial Association, 2016](#)). This paper interprets that purpose to mean that when the location of long-term financial risk changes, the scope of actuarial expertise should change with it. Extending analysis developed to protect the soundness of DB sponsors towards DC member outcomes, default design, and retirement-income adequacy is not a departure from pension actuarial work; it is a natural adaptation of its purpose to the contemporary pension structure.

Accumulation shapes the feasible set of retirement choices. Decumulation design is important, but the range of retirement choices first depends on the assets accumulated by retirement. Contributions and investment outcomes during accumulation determine the feasible levels of consumption, bequest, and annuitisation. The accumulation phase therefore forms the feasible set, while the decumulation phase selects a consumption, bequest, and annuitisation path within that set. DC accumulation must consequently manage not only expected terminal wealth, but also downside tails, variance, skewness, and the risk remaining at intermediate dates.

1.2 Practical problem and research gaps

DC contribution rates are generally specified by plan rules, whereas the assets available at retirement depend on investment performance. The economically relevant object is therefore not the one-period return itself but the distribution of retirement wealth generated over a long contribution horizon. Plan designers must explain and manage expected terminal wealth, variance, lower quantiles, distributional skewness, and the risk remaining after an intermediate account balance has been realised. A terminal-wealth mean–variance criterion in monetary units is thus a transparent and useful starting point.

The literature reviewed in Section 2.1 establishes lifecycle allocation with background wealth, state-dependent target-based DC strategies, and the distinct PCMV, DOMV, and TCMV notions of optimality for

time-inconsistent dynamic mean–variance problems. These strands provide a theoretical basis for deriving controls from objectives, plan constraints, and member states rather than prescribing glide paths by age alone. Four gaps remain when the theory is translated into DC practice.

- Gap 1 The four solution concepts have not been organised under a common DC constraint.** The present value of future contributions and DB benefits affects risk-bearing capacity as background wealth, but it cannot be sold to finance investment inside the DC account. Total pension wealth must therefore be distinguished from investable account wealth, and PCMV, DOMV, cTCMV, and dTCMV must be characterised under the common state-dependent constraint $0 \leq \pi_t \leq X_t$. Existing studies do not provide a unified comparison of their optimality or equilibrium properties, free boundaries, and state-dependent controls under the same DC plan conditions.
- Gap 2 The difference between clipping and directly constrained feedback has not been traced through to member outcomes.** Clipping an unconstrained solution to its feasible range is inexpensive to implement. For example, [Ferreira Morici and Vigna \(2025\)](#) clip an unconstrained strategy to the investment-share interval $[0, 1]$ in view of the complexity of jointly modelling investment and additional contributions in a constrained DC accumulation problem. Clipping, however, projects only the current control while retaining the continuation value of the unconstrained problem. It therefore need not equal a solution that internalises the constraint in the future value function or equilibrium moments. How the gap varies by concept, date, account balance, and background wealth, and how far it propagates to glide paths and terminal and conditional distributions, remains unclear.
- Gap 3 Outcome evaluation after an intermediate balance has been realised is not adequately separated from time-zero evaluation.** Dynamic strategies are often compared through unconditional terminal distributions viewed at time zero. At a practical review date, however, the member’s current account balance is already observed. Past state differences should be separated from the prospective investment risk remaining from that state. A unified framework that reconstructs conditional terminal distributions at fixed dates and balances, and uses the law of total variance to separate within-state uncertainty from cross-state dispersion, is not yet standard across the four strategies.
- Gap 4 It is unclear to which solution concepts a skewness extension should be applied.** Mean–variance analysis is transparent but does not directly price asymmetry in terminal wealth. Equilibrium foundations for mean–variance–skewness (MVS) control exist ([Mu et al., 2019](#)), but it does not follow that MVS should be attached uniformly to every solution concept. The equality between cTCMVS and cTCMV in the unconstrained and corresponding clipped cases must be distinguished from the unresolved directly constrained case; dTCMV–MVS can alter the timing of risk-taking; and PCMV–MVS and DOMV–MVS can display strong non-concavity and discretisation sensitivity. A governance boundary is needed between main-paper findings and exploratory evidence.

These four gaps correspond one-to-one with objectives O1–O4 below: O1 addresses Gap 1, O2 Gap 2, O3 Gap 3, and O4 Gap 4. This mapping separates theoretical construction, approximation assessment, dynamic outcome evaluation, and the MVS extension into distinct research tasks.

1.3 Research objectives and analytical design

The research objectives are defined as O1–O4, in direct correspondence with the four gaps. O1 constructs the theory under the common DC constraint; O2 evaluates clipping error; O3 develops conditional outcome evaluation after an intermediate balance is observed; and O4 examines selective MVS extensions. Table 1 states what each objective establishes and where it is tested.

Table 1: Research objectives and principal validation sections

Objective	What this paper establishes	Main validation sections
O1	Characterise the state-dependent PCMV, DOMV, cTCMV, and dTCMV controls under $0 \leq \pi_t \leq X_t$, and verify their respective optimality or equilibrium properties.	Section 4 and Appendix B
O2	Distinguish clipped approximations from directly constrained feedback and measure their effects on policy shape, binding regions, and terminal and conditional distributions.	Sections 4–6
O3	Introduce a rolling conditional outcome profile based on the law of total variance and visualise expected outcomes, lower and upper tail risk, and current investment by realised state.	Sections 6 and 8
O4	Organise MVS by solution concept; separate the cTCMVS equality for the unconstrained and clipped cases from the unresolved directly constrained problem; and assess dTCMV–MVS coefficient sensitivity and the numerical and governance limitations of PCMV–MVS and DOMV–MVS.	Section 7

O1 and O2 identify the theoretical solutions under the DC constraint and the circumstances in which a low-cost approximation is adequate. O3 evaluates each policy not only through its time-zero terminal distribution but also through the remaining distribution conditional on an observed intermediate balance. O4 distinguishes

the meaning of skewness under each concept and evaluates implementability in light of numerical sensitivity. Contributions C1–C6, stated at the end of Section 2, are the theoretical, computational, and practical outputs obtained by pursuing these objectives.

The four solution concepts are summarised below for later reference; formal definitions are given in Table 4 and Section 3.3.

Guide to the four solution concepts

Concept	Decision principle	Intended practical role
PCMV	Commit at time zero to a terminal target and policy.	Design benchmark
DOMV	Recompute the PCMV problem at each review date.	Periodic review
cTCMV	Seek a time-consistent equilibrium among successive selves under constant variance aversion.	Standard default
dTCMV	Seek a time-consistent equilibrium under total-pension-wealth-dependent variance aversion.	Personalised advice based on total pension wealth

Throughout the paper, risky-asset investment above the DC balance, short selling, and borrowing against future contributions are prohibited. The formal constraint and the present value of future contributions and deterministic DB benefits are introduced in Section 3. Background wealth can enter the objective or variance-aversion specification, but the investable amount is always limited to the DC account balance.

Verification conditions and free boundaries are developed in Section 4. The implementation comparison distinguishes the unconstrained solution, its ex-post clipped projection, and the directly constrained feedback that internalises the constraint from the outset. “Unconstrained solution” includes semi-analytical controls, such as dTCMV coefficients obtained numerically from an integral equation, provided that the resulting control remains easy to implement. Formal definitions are collected in Section 4.1.

The numerical analysis uses a common Markov–Gauss–Hermite transition kernel for backward solution, forward distribution propagation, terminal evaluation, and rolling conditional evaluation. Differences can therefore be attributed to the policies rather than to different integration or distribution-evaluation methods. A constant-proportion strategy (CP) is included as the simplest benchmark. MVS is not treated as a uniform additional operating layer: the cTCMVS equality in the unconstrained and clipped cases, the unresolved directly constrained case, dTCMV–MVS coefficient sensitivity, and exploratory PCMV–MVS and DOMV–MVS calculations are kept distinct.

The relationship between the literature and objectives O1–O4 is summarised in Table 2.

2 Literature Review and Positioning

The purpose of this section is not to repeat the practical motivation, but to distinguish established results from the incremental contribution of this paper. The literature is organised into three foundations: dynamic investment theory; constraints and implementation approximations; and computation, dynamic outcome evaluation, and higher moments. Table 2 serves as a map for Sections 2.1–2.3.

Table 2: Literature foundations, remaining gaps, and this paper’s response

Foundation	Established knowledge	Gap addressed here	Objective
Dynamic investment theory (Section 2.1)	Background wealth, target-based DC, and PCMV, DOMV, and TCMV provide state-dependent allocation and distinct notions of optimality.	Distinguish non-tradable pension wealth from the DC investment limit and characterise all four concepts under one constraint.	O1
Constraints and implementation approximations (Section 2.2)	Convex constraints create free boundaries, and clipping offers a low-cost implementation approximation.	Measure the clipped-versus-directly-constrained gap consistently across state regions, glide paths, and distributional outcomes.	O2
Computation, outcome evaluation, and MVS (Section 2.3)	Markov methods, quadrature, conditional-distribution analysis, and MVS equilibrium control provide established foundations.	Integrate a common transition kernel, separation of remaining-horizon outcomes, and solution-concept-specific MVS governance.	O3–O4

2.1 Dynamic investment theory: background wealth, target-based DC, and time inconsistency

Lifecycle portfolio theory treats labour income and human capital as non-tradable background wealth and derives dynamic allocation from total risk-bearing capacity (Merton, 1969; Bodie et al., 1992; Viceira, 2001; Cocco et al., 2005). DC research has shown that a monotonically declining age-only glide path need not dominate and that state-dependent design should account for balances, contributions, and targets (Cairns et al., 2006; Basu et al., 2011; Forsyth and Vetzal, 2019; Estrada, 2019). Target-based DC research developed by Vigna and co-authors connects replacement-rate targets, downside risk, quadratic loss, and the mean–variance efficient frontier (Vigna and Haberman, 2001; Haberman and Vigna, 2002; Højgaard and Vigna, 2007; Vigna, 2014; Menoncin and Vigna, 2017).

In dynamic mean–variance problems, PCMV commits to the initial policy (Li and Ng, 2000; Zhou and Li, 2000); DOMV repeatedly solves the remaining-horizon problem (Pedersen and Peskir, 2017; Menoncin

and Vigna, 2020); and TCMV seeks an equilibrium among future selves (Basak and Chabakauri, 2010; Bjork and Murgoci, 2014; Bjork et al., 2017). Each is a legitimate but different notion of optimality. The closest comparative study, van Staden et al. (2021), compares time-zero terminal distributions for PCMV, DOMV, constant- and wealth-dependent TCMV, and constant-proportion strategies at a common mean. It documents the distinctive PCMV distribution and partial first-order stochastic dominance by a constant-proportion strategy in the low-terminal-wealth region. This paper adds deterministic contributions, non-tradable pension wealth, and a DC-specific investment ceiling, and evaluates conditional outcomes after an intermediate balance has been observed.

2.2 Constrained control and clipped approximations

Short-sale prohibitions, bankruptcy constraints, and convex portfolio sets generate constrained value functions, projection structures, and free boundaries (Li et al., 2002; Li and Xu, 2015; Bensoussan et al., 2014; Guan et al., 2022). When both contributions and investment constraints are introduced into DC models, direct solution can become complex, and clipping an unconstrained control to the feasible set is used as a practical alternative (Di Giacinto et al., 2014; Ferreira Morici and Vigna, 2025).

Clipping satisfies the pointwise constraint but retains the continuation value of the unconstrained problem. It therefore need not coincide with a solution that internalises the constraint in future value functions or equilibrium moments. The unresolved question is how this difference varies across solution concepts, dates, balances, and background wealth, and how far it propagates into glide paths and member outcomes. This is the central gap addressed by O2.

2.3 Computation, dynamic outcome evaluation, and MVS

Markov-chain approximation, semi-Lagrangian schemes, and Gauss–Hermite quadrature are established numerical methods for stochastic control and normal-shock integration (Kushner and Dupuis, 2001; Falcone and Ferretti, 2014; Abramowitz and Stegun, 1972; Judd, 1998). This paper does not claim novelty for the individual numerical methods. Its computational contribution is to use one quadrature-induced transition kernel across the four constrained mean–variance systems, comparisons with clipped controls, forward propagation of terminal and rolling distributions, and the nested MV-to-MVS validation.

Rolling in DOMV is a *decision principle*: the remaining-horizon problem is solved again. The rolling conditional evaluation introduced here is instead an *outcome-evaluation principle*: for any policy, it reconstructs the conditional distribution from an intermediate date and balance. The law of total variance separates remaining-horizon risk from cross-state dispersion embedded in time-zero terminal variance.

For MVS, we build on the equilibrium formulation of Mu et al. (2019). In their constant-coefficient unconstrained Black–Scholes case, cTCMVS and cTCMV generate the same equilibrium risky investment, and the clipped controls constructed from that common unconstrained solution also coincide. The equality does not follow for a directly constrained cTCMVS problem in which the state-dependent constraint is internalised in the continuation value, and it is not proved here. dTCMV–MVS can generate coefficient-dependent glide paths, whereas PCMV–MVS and DOMV–MVS are kept as exploratory results because of non-concavity and numerical sensitivity.

The six contributions below are analytical outputs rather than restatements of the objectives. C1–C2 concern theory and optimality, C3–C5 computation and outcome evaluation, and C6 practical translation.

- C1 Characterisation of four solution concepts under a common DC constraint.** PCMV, DOMV, cTCMV, and dTCMV are formulated under $0 \leq \pi_t \leq X_t$, with their respective optimality or equilibrium properties expressed through standard or extended HJB systems and verification theorems. This primarily addresses O1.
- C2 Separation and error assessment of clipping and directly constrained feedback.** The unconstrained solution, clipped projection, and directly constrained feedback are distinguished, and their differences are propagated through free boundaries, policy shapes, glide paths, terminal distributions, and conditional distributions. This identifies where clipping is adequate and where direct constrained optimisation is required. This primarily addresses O2.
- C3 A common backward–forward computational engine.** A Markov–Gauss–Hermite transition kernel is shared across backward policy solution, forward distribution propagation, terminal evaluation, and the nested MVS validation. This separates policy differences from numerical-integration differences and supports O1–O4.
- C4 A rolling conditional outcome profile.** For any policy, the conditional terminal distribution is reconstructed at a fixed intermediate balance. Expected remaining-horizon outcomes, lower quantiles, upside participation, variance, and current investment are evaluated by state, and the law of total variance separates within-state uncertainty from cross-state dispersion. This primarily addresses O3.
- C5 Integrated internal consistency, external reproduction, and MVS diagnostics.** Moment consistency, probability-mass conservation, grid and quadrature sensitivity, and external reproduction are tested within the same computational structure. MVS implementations are checked to return to MV when the third-moment coefficient is zero; the cTCMVS equality is scoped to the unconstrained

and clipped cases; and coefficient-dependent dTCMV–MVS results and the non-concavity of PCMV–MVS and DOMV–MVS are documented. This primarily addresses O4 and supports the numerical credibility of O1–O3.

C6 Translation into a three-layer operating design. PCMV is used as a target-setting and plan-design benchmark; cTCMV as the standard default; DOMV for periodic review; and dTCMV for personalised advice based on total pension wealth. MVS is retained as a research extension until coefficient explainability, non-concavity, grid convergence, and independent reproduction are sufficiently resolved. This primarily addresses O2–O4.

2.4 Evaluation criteria and structure of the paper

Table 3 summarises five criteria used to assess the paper’s results: relevance, quality, originality, value, and future development. It provides a guide to where practical need, theoretical and numerical reliability, and differences from the literature can be verified.

Table 3: Five criteria for assessing the paper’s contribution

Criterion	Evidence in this paper
Relevance	Standard DC defaults, periodic review, personalised advice based on total pension wealth, and the applicability of low-cost clipping are addressed in one framework.
Quality	Constrained HJB and extended-HJB systems, verification theorems, a common backward–forward engine, internal diagnostics, and external reproduction are presented, while completed validation is distinguished from future convergence work.
Originality	Four dynamic mean–variance concepts are compared under a common DC constraint; clipping error is propagated to distributional outcomes; and a rolling conditional outcome profile is introduced.
Value	Mathematical differences are translated into a PCMV design benchmark, a cTCMV–DOMV–dTCMV three-layer operating model, and state-specific outcome communication.
Future	Further work is specified for grid stability, non-concave MVS branches, stochastic income and mortality, and decumulation.

Sections 3–4 present the model and constrained theory; Section 5 the computational engine; Section 6 the numerical results; Section 7 the MVS extensions; Section 8 the actuarial interpretation; and Section 9 limitations and conclusions. Reproducibility code, principal numerical data, and supplementary figures are provided through the English repository described in Appendix A.

3 Model and Solution Concepts

3.1 Financial market and the DC account

Consider a finite horizon $[0, T]$. The risk-free asset B_t and risky asset S_t satisfy

$$\frac{dB_t}{B_t} = r dt, \quad (1)$$

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t, \quad \sigma > 0, \quad (2)$$

where r, μ, σ are constant and $\beta = \mu - r$ is the excess return. Let X_t denote the DC account balance, π_t the amount invested in the risky asset, and $c_t \geq 0$ a deterministic contribution flow. Then

$$dX_t = (rX_t + c_t + \beta\pi_t) dt + \sigma\pi_t dW_t, \quad X_0 = x_0 \geq 0. \quad (3)$$

For a standard DC default, it is natural to prohibit both short selling and risky investment in excess of the account balance. We therefore impose

$$0 \leq \pi_t \leq X_t. \quad (4)$$

This is part of the control set itself, not a numerical truncation imposed after solving an unconstrained problem.

3.2 Background wealth and an auxiliary transformation

Non-tradable pension-related cash flows, such as future contributions and a deterministic DB benefit, are represented by their present value

$$H_t = \int_t^T e^{-r(s-t)} c_s ds + e^{-r(T-t)} D_T, \quad (5)$$

where D_T is a deterministic terminal benefit. The notation “DB benefit” is used for brevity, but the same formulation covers other deterministic retirement benefits. If $D_T = 0$, background wealth consists only of future contributions. The process satisfies

$$\dot{H}_t = rH_t - c_t, \quad H_T = D_T. \quad (6)$$

Define total pension wealth

$$Y_t = X_t + H_t. \quad (7)$$

Using (3) and (6),

$$dY_t = (rY_t + \beta\pi_t) dt + \sigma\pi_t dW_t, \quad Y_T = X_T + D_T. \quad (8)$$

The transformation serves two purposes. First, $X_t + H_t$ is interpreted as total pension wealth and enters the wealth-dependent variance-aversion specification of dTCMV. Second, after removing the investment constraint, it converts the contribution model into a standard Black–Scholes-type auxiliary problem from which inexpensive unconstrained benchmarks can be constructed. Importantly, Y_t is not tradable wealth. The investable amount remains X_t . Accordingly, the directly constrained HJB systems, verification results, and baseline numerical solution are formulated in X -space; Y -space is used only for interpretation and unconstrained benchmarks.

3.3 Admissible controls and glide paths

Definition 1 (Admissible control in X -space). For an initial state (t, x) , define

$$\mathcal{A}_{t,x}^X = \{\pi : \pi_s \text{ is progressively measurable, } 0 \leq \pi_s \leq X_s^\pi, \quad X_s^\pi \geq 0, \quad s \in [t, T]\}. \quad (9)$$

The glide path is the risky-asset share within the DC account,

$$g_t = \frac{\pi_t}{X_t}. \quad (10)$$

When $X_t = 0$, only $\pi_t = 0$ is admissible and the ratio is undefined. Numerically, $x = 0$ is the lower grid boundary and negative discrete transition values are floored at zero. Although $\pi = 0$ at $x = 0$, the state is not absorbing because contributions remain positive. Positive low balances are evaluated by linear interpolation on a non-uniform grid refined near zero.

Table 4: Guide to the four mean–variance solution concepts

Concept	Decision principle	Time property	Main input	Intended practical role
PCMV	Commit at time zero to a terminal target and policy	Pre-commitment	Embedded target m	Design benchmark for target setting and the mean–variance trade-off
DOMV	Re-solve the remaining-horizon problem at every reached date and state	Rolling optimality	λ_D or a local target	Periodic review
cTCMV	Seek an intra-personal equilibrium immune to profitable infinitesimal deviations by the current self	Subgame-perfect-Nash-type equilibrium	Constant γ_c	Standard default
dTCMV	Seek an intra-personal equilibrium with variance aversion determined by total pension wealth	Subgame-perfect-Nash-type equilibrium	$\Gamma(t, x) = \rho_d/(x + H_t)$	Personalised advice based on total pension wealth

4 Constrained Mean–Variance Theory

4.1 Three objects of comparison: unconstrained solution, clipped approximation, and directly constrained feedback

4.1.1 Why the three objects must be distinguished

The practical question is not whether an unconstrained solution is “better” than a constrained one. An unconstrained policy may violate the DC constraint and is therefore not directly implementable. Its role is to provide a low-cost input for a clipped approximation. The relevant question is how closely that clipped approximation reproduces feedback from a problem that internalises the constraint in the continuation value.

Let $\mathcal{J} := \{\text{PCMV}, \text{DOMV}, \text{cTCMV}, \text{dTCMV}\}$. For $j \in \mathcal{J}$, define

$$\begin{cases} \pi^{j,\text{clip}}(t, x) := \text{Proj}_{[0,x]} \{\pi^{j,0}(t, x + H_t)\} & (\text{clipped approximation}), \\ \pi^{j,\text{str}}(t, x) & (\text{directly constrained feedback}). \end{cases} \quad (11)$$

The first is a projection of a control from a different, unconstrained problem; the second reflects the effect of future binding constraints on the constrained value function or equilibrium moments. Equation (11) therefore does not assert that clipping an unconstrained policy produces the directly constrained solution.

4.1.2 Unconstrained solutions in Y -space

Because $Y_t = X_t + H_t$ satisfies

$$dY_t = (rY_t + \beta\pi_t) dt + \sigma\pi_t dW_t,$$

the unconstrained auxiliary problems take a standard Black–Scholes form. Let $A := \beta^2/\sigma^2$ and $y = x + H_t$. The unconstrained controls used to construct the clipped approximations are

$$\pi^{p,0}(t, y; m) = \frac{\beta}{\sigma^2} \{m e^{-r(T-t)} - y\}, \quad (12)$$

$$\pi^{D,0}(t, y; \lambda_D) = \frac{\beta}{2\lambda_D\sigma^2} e^{(A-r)(T-t)}, \quad (13)$$

$$\pi^{c,0}(t, y) = \frac{\beta}{\gamma_c\sigma^2} e^{-r(T-t)}, \quad (14)$$

$$\pi^{d,0}(t, y) = \theta_{\rho_d}(t)y. \quad (15)$$

The PCMV formula depends on the embedded target m ; the DOMV formula on the scalarisation parameter λ_D . In the numerical comparison, DOMV is calibrated through a parameter or local target consistent with its objective. The DOMV parameter λ_D and dTCMV scale parameter ρ_d serve different functions and are kept notationally distinct. The time-varying ratio θ_{ρ_d} is the unique solution of a Volterra-type integral equation associated with the Björk–Murgoci wealth-dependent variance-aversion problem. Numerically, the equation is converted into a backward ODE system with two auxiliary integrals. These expressions correspond to the unconstrained formulas of [van Staden et al. \(2021\)](#), with their wealth variable interpreted as Y .

Definition 2 (Clipped approximation). For solution concept j , define

$$\pi^{j,\text{clip}}(t, x) = \text{Proj}_{[0,x]} \{ \pi^{j,0}(t, x + H_t) \}. \quad (16)$$

This policy is feasible, but it clips only the current control while retaining the continuation value of the unconstrained problem. It need not solve the corresponding constrained HJB or extended-HJB system.

Definition 3 (Directly constrained feedback). The feedback $\pi^{j,\text{str}}(t, x)$ is obtained by inserting $0 \leq \pi \leq x$ directly into the optimisation set of the HJB equation, extended-HJB system, or Markov–Gauss–Hermite backward recursion. Future binding of the constraint is therefore reflected in the constrained value function or equilibrium moments.

Remark 1 (Local projection versus ex-post clipping). Directly constrained feedback also has a local projection form, but its projection centre is calculated from the *constrained* value function or equilibrium moments. The clipped approximation projects the solution of a *different unconstrained problem*. In general,

$$\pi^{j,\text{str}}(t, x) \neq \pi^{j,\text{clip}}(t, x).$$

Equality requires either that the unconstrained policy remains feasible throughout the reachable region, or that the clipped candidate can separately be verified to satisfy the relevant constrained HJB system. The numerical analysis measures this distinction as the projection gap.

4.2 PCMV and DOMV: constrained quadratic-loss HJB problems

4.2.1 Embedded quadratic-loss problem

To maintain comparability with cTCMV and dTCMV, PCMV and DOMV evaluate total terminal pension wealth $X_T + D_T$. Since D_T is deterministic and unaffected by the control,

$$\text{Var}_{t,x}(X_T^\pi + D_T) = \text{Var}_{t,x}(X_T^\pi), \quad \mathbb{E}_{t,x}[X_T^\pi + D_T] = \mathbb{E}_{t,x}[X_T^\pi] + D_T, \quad (17)$$

and

$$(X_T^\pi + D_T - m)^2 = (X_T^\pi - (m - D_T))^2. \quad (18)$$

Thus, after shifting the target by D_T , the optimal PCMV and DOMV controls are the same as when $D_T = 0$. The benefit is retained only to keep the outcome unit common across the four concepts.

Definition 4 (PCMV). PCMV commits at the initial state to a mean–variance-efficient strategy and does not re-solve the original problem later. Following [van Staden et al. \(2021\)](#); [Li and Ng \(2000\)](#); [Zhou and Li \(2000\)](#) and the target-based DC literature ([Højgaard and Vigna, 2007](#); [Vigna, 2014](#); [Menoncin and Vigna, 2017](#)), the Li–Ng–Zhou embedding is written, for fixed $m \in \mathbb{R}$, as

$$L^m(t, x) = \inf_{\pi \in \mathcal{A}_{t,x}^X} \mathbb{E}_{t,x} [(X_T^\pi + D_T - m)^2]. \quad (19)$$

The feedback selected at time zero is then continued until maturity. Its relationship to mean-constrained PCMV is given in [Theorem 2](#).

The constrained HJB equation is

$$v_t^m + \inf_{0 \leq \pi \leq x} \left\{ (rx + c_t + \beta\pi)v_x^m + \frac{1}{2}\sigma^2\pi^2 v_{xx}^m \right\} = 0, \quad v^m(T, x) = (x + D_T - m)^2. \quad (20)$$

Assumption 1 (Regularity for verification). *Candidate functions are assumed to be $C^{1,2}$, to satisfy their terminal conditions continuously, and to have polynomial growth. The control processes satisfy the integrability, localisation, martingale, and limiting conditions needed to apply Itô's formula and pass to the terminal limit. Numerical solutions may more properly be interpreted as viscosity solutions; the classical framework is used here to make the projection structure explicit.*

For PCMV and DOMV, the control ratio $u_t = \pi_t/X_t \in [0, 1]$ converts the state-dependent interval into a fixed compact control set. Deterministic contributions are an affine drift term and do not obstruct standard weak-existence, moment-estimate, dynamic-programming, or viscosity-solution arguments. Proposition 3 in Appendix B records weak existence for the fixed-target value function, relaxed Markov controls, the outer PCMV target, and measurable DOMV target selection. Classical $C^{1,2}$ regularity, regular free boundaries, and a unique continuous strict feedback require additional work.

For cTCMV and dTCMV, existence of a global equilibrium is itself a fixed-point problem specific to time-inconsistent control. Existing one-sided short-sale results do not directly establish equilibrium existence under the state-dependent two-sided constraint $0 \leq \pi_t \leq X_t$. Theorems 5 and 7 are therefore local verification results conditional on the existence of sufficiently regular global candidates.

Theorem 1 (Verification of the constrained embedded PCMV problem). *Fix $m \in \mathbb{R}$. Suppose v^m satisfies Assumption 1 and HJB equation (20), with $v_{xx}^m > 0$. Then $v^m = L^m$. If*

$$\pi^{m,*}(t, x) = \text{Proj}_{[0, x]} \left(-\frac{\beta v_x^m(t, x)}{\sigma^2 v_{xx}^m(t, x)} \right) \quad (21)$$

is admissible and generates a unique state process, it is optimal for (19).

Proof. Apply Itô's formula to any admissible control π up to a localisation time τ_k :

$$\mathbb{E}[v^m(\tau_k, X_{\tau_k}^\pi)] - v^m(t, x) = \mathbb{E} \int_t^{\tau_k} (v_s^m + \mathcal{L}^{\pi_s} v^m)(s, X_s^\pi) ds \geq 0.$$

The inequality follows from the HJB minimisation. Passing to the terminal limit shows that v^m is a lower bound for every admissible quadratic loss. Since $v_{xx}^m > 0$, the Hamiltonian is strictly convex in π and its constrained minimiser is the projection in (21). Under that control the inequality becomes equality, proving optimality. $\square$

Corollary 1 (PCMV free-boundary regions). *Let*

$$u^m(t, x) = -\frac{\beta v_x^m(t, x)}{\sigma^2 v_{xx}^m(t, x)}.$$

The state space is divided into

$$\mathcal{R}_0^m = \{u^m \leq 0\}, \quad \mathcal{R}_I^m = \{0 < u^m < x\}, \quad \mathcal{R}_U^m = \{u^m \geq x\},$$

where

$$\begin{aligned} \mathcal{R}_0^m : \quad & v_t^m + (rx + c_t)v_x^m = 0, \\ \mathcal{R}_I^m : \quad & v_t^m + (rx + c_t)v_x^m - \frac{\beta^2}{2\sigma^2} \frac{(v_x^m)^2}{v_{xx}^m} = 0, \\ \mathcal{R}_U^m : \quad & v_t^m + \{(r + \beta)x + c_t\}v_x^m + \frac{1}{2}\sigma^2 x^2 v_{xx}^m = 0. \end{aligned}$$

The boundaries between these regions are free boundaries.

Proof. The projected control equals 0, u^m , or x according as the unconstrained centre lies below, inside, or above $[0, x]$. Substitution into the HJB equation yields the three PDEs. Because the switching conditions depend on derivatives of the unknown value function, the boundaries are endogenous. $\square$

4.2.2 From embedded quadratic loss to PCMV

The embedding follows the target-based DC literature connecting the mean-variance frontier and quadratic-loss objectives (Højgaard and Vigna, 2007; Vigna, 2014; Menoncin and Vigna, 2017). Define mean-constrained PCMV by

$$\inf_{\pi \in \mathcal{A}_{0, x_0}^X} \text{Var}(X_T^\pi + D_T) \quad \text{subject to} \quad \mathbb{E}[X_T^\pi + D_T] = d. \quad (22)$$

With Lagrange multiplier λ and $m = d + \lambda$, any feasible π satisfies

$$\mathbb{E}[(X_T^\pi + D_T - m)^2] = \text{Var}(X_T^\pi + D_T) + \lambda^2.$$

Theorem 2 (Verification of constrained PCMV). *Suppose that, for some $m = d + \lambda$, the conditions of Theorem 1 hold and its optimal control $\pi^{m,*}$ satisfies*

$$\mathbb{E}_{0,x_0}[X_T^{\pi^{m,*}} + D_T] = d.$$

Then $\pi^{m,}$ solves the mean-constrained PCMV problem (22).*

Proof. For every feasible control the embedded loss equals terminal variance plus the common constant λ^2 . Since $\pi^{m,*}$ minimises the embedded loss and satisfies the same mean constraint, cancelling λ^2 proves minimum variance. $\square$

4.2.3 DOMV as a rolling local PCMV problem

Definition 5 (DOMV). Following Pedersen and Peskir (2017); Menoncin and Vigna (2020); van Staden et al. (2021), DOMV treats every reached state (t, x) as a new initial condition, solves the remaining-horizon mean-variance problem, implements only the current component of the resulting optimal continuation policy, and repeats the operation at the next date. In the standard scalarised formulation,

$$\pi_{t,x}^{D,*} \in \arg \max_{\pi \in \mathcal{A}_{t,x}^X} \{ \mathbb{E}_{t,x}[X_T^\pi + D_T] - \lambda_D \text{Var}_{t,x}(X_T^\pi + D_T) \},$$

and $\pi^{\text{DOMV}}(t, x) = \pi_{t,x}^{D,*}(t, x)$. Equivalently, the implementation may use a local target mean $d(t, x)$ and embedded target $m(t, x) = d(t, x) + \lambda(t, x)$.

Theorem 3 (Verification of constrained DOMV). *For each (t, x) , suppose that the embedded problem with target $m(t, x)$ satisfies Theorem 1, with optimiser $\pi_{t,x}^{D,*}$ satisfying*

$$\mathbb{E}_{t,x}[X_T^{\pi_{t,x}^{D,*}} + D_T] = d(t, x).$$

Then $\pi_{t,x}^{D,}$ solves the local constrained mean-variance problem and*

$$\pi^{\text{DOMV}}(t, x) = \pi_{t,x}^{D,*}(t, x) \quad (23)$$

defines the rolling-optimal feedback.

Proof. At each fixed state, the remaining-horizon DOMV problem is an embedded PCMV problem with a state-specific target. Theorem 1 and the local mean-consistency condition give local optimality. DOMV implements only the current action and re-solves at the next state rather than committing to the whole continuation policy. $\square$

Remark 2 (DOMV and rolling evaluation). DOMV naturally represents strategy review after an intermediate balance has been observed. This decision principle is distinct from the rolling conditional outcome metric introduced later, which can be applied to any policy.

4.3 cTCMV: time-consistent equilibrium with constant variance aversion

4.3.1 Equilibrium criterion and moment functions

cTCMV is the time-consistent mean-variance equilibrium in which every temporal self uses the same constant variance-aversion coefficient $\gamma_c > 0$ (Basak and Chabakauri, 2010; Bjork et al., 2017; van Staden et al., 2021).

Definition 6 (cTCMV). At state (t, x) , define

$$J^c(t, x; \pi) = \mathbb{E}_{t,x}[X_T^\pi + D_T] - \frac{\gamma_c}{2} \text{Var}_{t,x}(X_T^\pi + D_T).$$

A candidate Markov feedback $\hat{\pi}^c$ is a constrained cTCMV equilibrium if no current self can obtain a positive first-order improvement by using an alternative admissible control $\nu \in [0, x]$ over a short interval $[t, t + h]$ and then returning to $\hat{\pi}^c$:

$$\liminf_{h \downarrow 0} \frac{J^c(t, x; \hat{\pi}^c) - J^c(t, x; \nu \otimes_h \hat{\pi}^c)}{h} \geq 0. \quad (24)$$

This is a subgame-perfect-Nash-type intra-personal equilibrium, not global re-optimisation of the remaining horizon as in DOMV.

Under the candidate equilibrium, define

$$F^c(t, x) = \mathbb{E}_{t,x}^{\hat{\pi}^c}[X_T + D_T], \quad V^c(t, x) = \mathbb{E}_{t,x}^{\hat{\pi}^c}[X_T + D_T] - \frac{\gamma_c}{2} \text{Var}_{t,x}^{\hat{\pi}^c}(X_T + D_T), \quad (25)$$

and, for control π ,

$$\mathcal{L}^\pi \phi = (rx + c_t + \beta\pi)\phi_x + \frac{1}{2}\sigma^2\pi^2\phi_{xx}. \quad (26)$$

The extended HJB system is

$$F_t^c + \mathcal{L}^{\hat{\pi}^c} F^c = 0, \quad F^c(T, x) = x + D_T, \quad (27)$$

$$V_t^c + \sup_{0 \leq \pi \leq x} \left\{ (rx + c_t + \beta\pi)V_x^c + \frac{1}{2}\sigma^2\pi^2V_{xx}^c - \frac{\gamma_c}{2}\sigma^2\pi^2(F_x^c)^2 \right\} = 0, \quad V^c(T, x) = x + D_T. \quad (28)$$

The final correction term accounts for the contribution of short-horizon shocks to the variance of the future conditional mean.

4.3.2 Local feedback representation

Theorem 4 (Local feedback for constrained cTCMV). *Suppose $(F^c, V^c, \hat{\pi}^c)$ satisfies (27)–(28) and*

$$D^c(t, x) = V_{xx}^c(t, x) - \gamma_c(F_x^c(t, x))^2 < 0. \quad (29)$$

Then the equilibrium feedback maximising the constrained Hamiltonian is

$$\hat{\pi}^c(t, x) = \text{Proj}_{[0, x]} \left(-\frac{\beta V_x^c(t, x)}{\sigma^2 D^c(t, x)} \right). \quad (30)$$

Proof. The control-dependent Hamiltonian is

$$h^c(\pi) = \beta V_x^c \pi + \frac{1}{2}\sigma^2 D^c \pi^2.$$

Since $D^c < 0$, it is strictly concave, and the constrained maximiser is the projection of its unconstrained maximiser onto $[0, x]$. $\square$

4.3.3 Verification theorem

Theorem 5 (Verification of constrained cTCMV). *Suppose $(F^c, V^c, \hat{\pi}^c)$ satisfies (27)–(28), $\hat{\pi}^c$ attains the pointwise supremum, and the regularity and integrability conditions required for the short-time expansion hold. Then $\hat{\pi}^c$ is a time-consistent constrained cTCMV equilibrium. If (29) holds, the equilibrium is represented by (30).*

Proof. Replace $\hat{\pi}^c$ by ν over $[t, t+h)$ and return to the candidate policy thereafter. Itô expansion and the backward equation for F^c yield

$$J^c(t, x; \nu \otimes_h \hat{\pi}^c) - J^c(t, x; \hat{\pi}^c) = h\{\mathcal{H}^c(t, x, \nu) - \mathcal{H}^c(t, x, \hat{\pi}^c)\} + o(h).$$

Pointwise maximisation makes the first-order term non-positive, which is exactly the equilibrium condition (24). $\square$

Corollary 2 (cTCMV free-boundary regions). *Let*

$$u^c(t, x) = -\frac{\beta V_x^c(t, x)}{\sigma^2\{V_{xx}^c(t, x) - \gamma_c(F_x^c(t, x))^2\}}.$$

The regions $\mathcal{R}_0^c = \{u^c \leq 0\}$, $\mathcal{R}_I^c = \{0 < u^c < x\}$, and $\mathcal{R}_U^c = \{u^c \geq x\}$ correspond respectively to controls 0, u^c , and x . Substitution into (27)–(28) gives the regional PDEs; their endogenous switching sets are free boundaries.

4.4 dTCMV: variance aversion dependent on total pension wealth

4.4.1 Variance aversion based on total pension wealth

dTCMV retains the same equilibrium concept as cTCMV but makes variance aversion depend on current total pension wealth. Following the wealth-dependent specification of [van Staden et al. \(2021\)](#), we set

$$\Gamma(t, x) = \frac{\rho_d}{x + H_t}, \quad \rho_d > 0. \quad (31)$$

The scale parameter ρ_d is calibrated once and remains fixed; the coefficient actually used by the current self is the state-dependent $\Gamma(t, x)$. For two members with the same DC balance, the member with greater present value of future contributions or DB benefits has lower variance aversion. dTCMV is therefore interpreted as personalised advice based on total pension wealth, not as a simple account-balance-proportional rule.

Remark 3 (Γ is not an investment limit). $\Gamma(t, x)$ scales variance aversion; it does not enlarge the amount that can be invested. The trading constraint remains $0 \leq \pi_t \leq X_t$. Background wealth affects risk-bearing capacity but cannot be invested inside the DC account.

4.4.2 Moment formulation

At state (t, x) , holding the current self's coefficient $\Gamma(t, x)$ fixed during a short deviation, define

$$J^d(t, x; \pi) = \mathbb{E}_{t,x}[X_T^\pi + D_T] - \frac{\Gamma(t, x)}{2} \text{Var}_{t,x}(X_T^\pi + D_T). \quad (32)$$

Definition 7 (dTCMV). dTCMV is the same Markov equilibrium as cTCMV, except that each self uses the total-pension-wealth-dependent coefficient $\Gamma(t, x)$. During a local deviation at (t, x) , that coefficient is treated as fixed. The equilibrium condition is

$$\liminf_{h \downarrow 0} \frac{J^d(t, x; \hat{\pi}^d) - J^d(t, x; \nu \otimes_h \hat{\pi}^d)}{h} \geq 0. \quad (33)$$

Under candidate feedback $\hat{\pi}^d$, define

$$F^d(t, x) = \mathbb{E}_{t,x}^{\hat{\pi}^d}[X_T + D_T], \quad G^d(t, x) = \mathbb{E}_{t,x}^{\hat{\pi}^d}[(X_T + D_T)^2]. \quad (34)$$

Then

$$F_t^d + \mathcal{L}^{\hat{\pi}^d} F^d = 0, \quad F^d(T, x) = x + D_T, \quad (35)$$

$$G_t^d + \mathcal{L}^{\hat{\pi}^d} G^d = 0, \quad G^d(T, x) = (x + D_T)^2. \quad (36)$$

Because $\Gamma(t, x)$ is state-dependent, differentiating the composite equilibrium value alone would introduce Γ_x and Γ_{xx} even though the current self holds $\Gamma(t, x)$ fixed while evaluating a deviation. The extended-HJB representation must therefore be written directly in terms of F^d and G^d . A first-order expansion of $F^d - (\Gamma/2)\{G^d - (F^d)^2\}$ gives the local Hamiltonian

$$\mathcal{H}^d(t, x, \pi) = (1 + \Gamma F^d)(F_t^d + \mathcal{L}^\pi F^d) - \frac{\Gamma}{2}(G_t^d + \mathcal{L}^\pi G^d). \quad (37)$$

This is the moment-form specialisation of the equilibrium framework of Bjork et al. (2014, 2017). The contribution of this paper is to solve the local equilibrium under the DC investment limit and derive its constrained projection and free-boundary structure.

4.4.3 Local feedback and verification

Theorem 6 (Local feedback for constrained dTCMV). Suppose $(F^d, G^d, \hat{\pi}^d)$ satisfies (35)–(36) and

$$\hat{\pi}^d(t, x) \in \arg \max_{0 \leq \pi \leq x} \mathcal{H}^d(t, x, \pi). \quad (38)$$

Let

$$A^d(t, x) = (1 + \Gamma F^d)F_x^d - \frac{\Gamma}{2}G_x^d, \quad (39)$$

$$B^d(t, x) = (1 + \Gamma F^d)F_{xx}^d - \frac{\Gamma}{2}G_{xx}^d. \quad (40)$$

If $B^d(t, x) < 0$, then

$$\hat{\pi}^d(t, x) = \text{Proj}_{[0, x]} \left(-\frac{\beta A^d(t, x)}{\sigma^2 B^d(t, x)} \right). \quad (41)$$

Proof. The control-dependent part of (37) is

$$h^d(\pi) = \beta A^d \pi + \frac{1}{2} \sigma^2 B^d \pi^2.$$

Since the current value of Γ is fixed during the deviation, derivatives of Γ do not appear in this local maximisation. For $B^d < 0$, the Hamiltonian is strictly concave and its constrained maximiser is the projected unconstrained maximiser. $\square$

Theorem 7 (Verification of constrained dTCMV). Suppose $(F^d, G^d, \hat{\pi}^d)$ satisfies (35)–(36), $\hat{\pi}^d$ maximises (37) over $[0, x]$ pointwise, and the regularity and integrability conditions for the short-time expansion hold. Then $\hat{\pi}^d$ is a constrained dTCMV equilibrium. If $B^d < 0$, it has representation (41).

Proof. For a short deviation ν over $[t, t+h]$, with $\Gamma(t, x)$ fixed, first-order Itô expansions of the conditional first and second moments give

$$J^d(t, x; \nu \otimes_h \hat{\pi}^d) - J^d(t, x; \hat{\pi}^d) = h\{\mathcal{H}^d(t, x, \nu) - \mathcal{H}^d(t, x, \hat{\pi}^d)\} + o(h).$$

Pointwise maximisation makes the first-order term non-positive, proving (33). $\square$

Corollary 3 (dTCMV free-boundary regions). *Let $u^d(t, x) = -\beta A^d(t, x)/(\sigma^2 B^d(t, x))$. The regions $\mathcal{R}_0^d = \{u^d \leq 0\}$, $\mathcal{R}_I^d = \{0 < u^d < x\}$, and $\mathcal{R}_U^d = \{u^d \geq x\}$ correspond to controls 0, u^d , and x . In each region, F^d and G^d satisfy the linear PDEs obtained by substituting the relevant control into (35)–(36); the switching sets are free boundaries.*

Remark 4 (Why dTCMV cannot be reduced to derivatives of V^d alone). If Γ were constant, the formulation could be reduced to a cTCMV-type value function. When $\Gamma(t, x)$ is state-dependent, derivatives of the composite value include derivatives of Γ , whereas the current self holds $\Gamma(t, x)$ fixed during a deviation. The equilibrium-consistent Hamiltonian must therefore retain the separate moment functions as in (37).

4.5 CP: constant-proportion benchmark

Unlike the four mean–variance concepts, a constant-proportion strategy is not derived from an optimality or equilibrium condition. It is a simple implementable benchmark.

Definition 8 (DC constant-proportion strategy). For $\theta_{cp} \in [0, 1]$, define

$$\pi^{cp}(t, x) = \theta_{cp}x, \quad g^{cp}(t, x) = \frac{\pi^{cp}(t, x)}{x} = \theta_{cp}, \quad (42)$$

with $\pi^{cp}(t, 0) = 0$.

CP automatically satisfies $0 \leq \pi \leq x$ but has no verification theorem or endogenous free boundary. We calibrate θ_{cp} so that expected terminal wealth matches the four MV strategies, following the equal-mean comparison principle of van Staden et al. (2021).

4.6 Unified exact projection and free-boundary structure

Although the objectives and notions of optimality differ, placing the DC constraint inside the optimisation gives each concept a common structural form: a local candidate computed from the constrained continuation object is projected onto $[0, x]$.

Theorem 8 (Unified projection structure). *Suppose candidate solutions exist and satisfy*

$$v_{xx}^m > 0, \quad D^c = V_{xx}^c - \gamma_c(F_x^c)^2 < 0, \quad B^d < 0.$$

Then

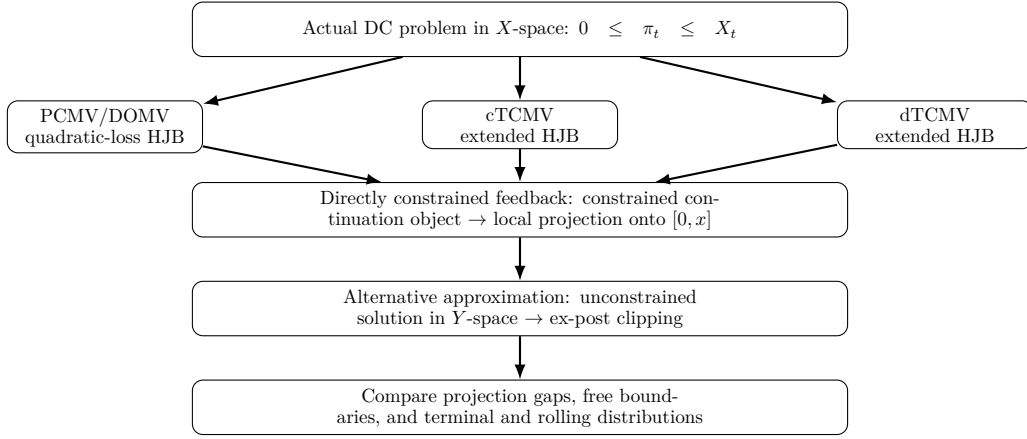
$$\begin{aligned} \pi^{p,\text{str}}(t, x) &= \text{Proj}_{[0,x]} \left(-\frac{\beta v_x^m}{\sigma^2 v_{xx}^m} \right), \\ \pi^{c,\text{str}}(t, x) &= \text{Proj}_{[0,x]} \left(-\frac{\beta V_x^c}{\sigma^2 D^c} \right), \\ \pi^{d,\text{str}}(t, x) &= \text{Proj}_{[0,x]} \left(-\frac{\beta A^d}{\sigma^2 B^d} \right). \end{aligned}$$

DOMV has the PCMV form applied to each local re-solved problem.

Proof. Apply Theorems 1, 4, and 6. The common feature is a quadratic local Hamiltonian whose curvature has the required sign. The projection centre, however, is constructed from a different constrained value or moment system for each concept. The commonality is therefore the form of the local projection, not equality with ex-post clipping of an unconstrained solution. $\square$

Remark 5 (Direct projection versus clipping). The quantities in Theorem 8 are derivatives of constrained HJB or extended-HJB solutions. They differ from the Y -space unconstrained controls used in Section 4.1. The latter provide inexpensive clipped approximations; the theorem characterises the directly constrained policies.

The state space is divided into lower-bound, interior, and upper-bound regions according as the local candidate lies below zero, inside $(0, x)$, or above x . Because the switching conditions depend on the unknown value or moment functions, the boundaries are endogenous free boundaries.



5 A Unified Markov–Gauss–Hermite Engine for Constrained MV Control

5.1 Positioning and contribution of the computational engine

To solve and evaluate the four constrained MV problems under common numerical conditions, we combine Gauss–Hermite conditional expectations, grid-based backward recursion, and deterministic forward propagation of probability mass. The term *unified Markov–Gauss–Hermite engine* refers to this computational architecture, not to a new quadrature rule. Standard Gauss–Hermite quadrature (Abramowitz and Stegun, 1972; Judd, 1998), Markov-chain and semi-Lagrangian methods for controlled diffusions (Kushner and Dupuis, 2001; Falcone and Ferretti, 2014), and numerical HJB methods for MV allocation (Wang and Forsyth, 2012a,b) are well established.

The contribution is the common use of the same one-step expectation evaluator and induced transition kernel across four solution concepts, directly constrained and clipped policies, terminal and rolling distributions, and MV/MVS nesting checks. Table 5 separates standard components from their integrated role in this paper.

Table 5: Positioning of the unified Markov–Gauss–Hermite engine

Component	Methodological status	Function in this paper
Gauss–Hermite conditional expectation	Standard deterministic quadrature for normal shocks	Uses one expectation evaluator for PCMV, DOMV, cTCMV, dTCMV, and MVS, keeping shock approximation and interpolation common.
Markov/semi-Lagrangian backward recursion	Established grid method for stochastic control	Inserts the state-dependent interval $[0, x]$ directly into each node’s admissible set and handles ordinary Bellman and extended-HJB moment recursions in one structure.
Parallel evaluation of direct constraints and clipping	Clipping is an inexpensive ex-post approximation, not the same problem	Compares both policies on the same state and control grids, quadrature nodes, interpolation, and forward evaluator.
Forward propagation with the same kernel	Standard Markov probability-mass propagation	Uses the backward kernel for terminal and rolling laws, CDFs, quantiles, tails, and moment diagnostics.
Common implementation of four MV concepts and MVS	Integrated design specific to this paper	Places different temporal decision principles and nested MVS extensions in one input/output and diagnostic framework.

The architecture has four practical benefits. First, it identifies policy differences because expectation rules, random seeds, and forward evaluators are held fixed. Second, it propagates locally verified feedback controls to terminal and conditional member outcomes. Third, deterministic quadrature avoids Monte Carlo noise in small projection gaps and free-boundary locations. Fourth, it supports numerical governance through backward–forward moment checks, mass conservation, and the $\eta = 0$ return from MVS to MV.

Scope of the numerical contribution. The claim is not a new “Gauss–Hermite method” but an integrated backward–forward architecture serving six linked purposes: four constrained MV concepts; direct-versus-clipped comparisons; MV/MVS nesting; backward policy solution; forward terminal and rolling evaluation; and implementation governance. The forward procedure is not pathwise random Monte Carlo. It deterministically transports probability mass along the discrete transition kernel induced by the quadrature nodes.

Why Gauss–Hermite quadrature is used. Conditional expectations must be evaluated repeatedly for every date, state, and candidate control. Gauss–Hermite quadrature avoids sampling noise in the inner loop. Reusing its nodes for forward propagation makes policy shapes, free boundaries, terminal laws, and rolling laws reproducible.

5.2 Numerical state space and controls

Discrete-index convention. Throughout this section, state-dependent functions retain the continuous-time ordering (t, x) after discretisation: for example, $V_n(x_i) := V(t_n, x_i)$, $M_n(x_i) := M(t_n, x_i)$, and $\pi_n(x_i) :=$

$\pi(t_n, x_i)$. The time index is therefore attached to the function, while x_i denotes the current wealth-grid point. Probability mass is stored as $p_{n,i}$, the control-grid index is a , the Gauss–Hermite node index is k , and the strategy index used in rolling evaluation is j . A Gauss–Hermite successor generated from (t_n, x_i) under candidate control π is denoted $\hat{x}_{n+1}^{(k)}(x_i, \pi)$.

5.2.1 Time grid

Partition $[0, T]$ into N_t steps,

$$t_n = n\Delta t, \quad \Delta t = \frac{T}{N_t}, \quad n = 0, 1, \dots, N_t. \quad (43)$$

The baseline model uses monthly decisions over 40 years: $N_t = 480$ and $\Delta t = 1/12$. This is a plan-frequency specification, not a proof of convergence to the continuous-time HJB solution. The broad sensitivity analysis uses the coarser screening grid $N_t = 80$ for computational economy and is not used as an accuracy certificate for the monthly baseline. Backward recursion proceeds from $N_t - 1$ to 0, and forward propagation from 0 to $N_t - 1$.

5.2.2 Wealth grid

Let

$$0 = x_0 < x_1 < \dots < x_I = x_{\max}, \quad (44)$$

with $N_x = I + 1$. Because the constraint is most influential at low balances, we use the non-uniform grid

$$x_i = x_{\max} \left(\frac{i}{I} \right)^\kappa, \quad \kappa > 1, \quad (45)$$

and explicitly insert the deterministic initial balance $x_0^{\text{init}} = 1/12$ as a grid point. The sensitivity screening uses $(I + 1, \kappa) = (101, 2.5)$, while the monthly baseline uses 151 wealth nodes.

Refinement near zero is economically important. The admissible interval $[0, x]$ collapses as $x \downarrow 0$, so a small error in investment amount can become a large error in the ratio $g = \pi/x$. At $x = 0$, all ratio candidates imply the same amount $\pi = 0$, and an arbitrary tie-breaking rule can contaminate interpolation. Moreover, dTCMV has state-dependent $\Gamma(t, x) = \rho_d/(x + H_t)$, so low-balance errors propagate through the interaction of background wealth and the upper constraint. We therefore combine a refined low-wealth grid, explicit inclusion of the initial balance, and interpolation of the amount $\pi = xg$ rather than the ratio.

All PDE and distribution calculations remain in X -space. For dTCMV, only the coefficient

$$\Gamma(t_n, x_i) = \frac{\rho_d}{x_i + H_{t_n}} \quad (46)$$

is evaluated using background wealth. Since H_{t_n} is deterministic, no extra state dimension or Y -space grid is required.

5.2.3 Control grid and admissible interval

At node (t_n, x_i) ,

$$\Pi_{n,i} = [0, x_i]. \quad (47)$$

A discrete candidate set is

$$\pi_i^{(a)} = \frac{a}{N_a - 1} x_i, \quad a = 0, 1, \dots, N_a - 1. \quad (48)$$

A coarse search can be followed by one-dimensional golden-section or Brent refinement over $[0, x_i]$.

Remark 6 (Control search and the projection formula). The theory expresses smooth constrained feedback as a projection of a local unconstrained candidate. The implementation need not rely on derivative estimates: direct search over $[0, x_i]$ treats interior and binding regions uniformly. The projection formula is used as a diagnostic and interpretive device for free boundaries and projection gaps.

5.3 Gauss–Hermite conditional expectations

By this point, the analytical structure is largely explicit: the unconstrained controls have closed forms, and the directly constrained problems have been reduced to state-dependent feedback rules and binding regions. What remains numerically is the evaluation of the HJB or extended-HJB continuation equations. In preliminary implementations, solving those PDEs directly by finite-difference methods proved noticeably sensitive to the time–state mesh and to derivative discretisation, especially near switching boundaries. Even when a computed surface appeared smooth, it was difficult to tell whether a local feature reflected the underlying control problem or a mesh artefact. This numerical ambiguity motivated a different primary engine for the results reported below.

Accordingly, the main computation evaluates one-step conditional expectations directly and builds the dynamic recursion from the induced transition kernel. Gauss–Hermite quadrature is used for the normal shock, the same kernel is used in the backward recursion and in forward probability-mass propagation, and numerical credibility is assessed through grid refinement, boundary-mass diagnostics, and backward–forward moment consistency. This choice does not eliminate discretisation error; rather, it replaces direct finite-difference sensitivity of PDE derivatives with transparent time-, state-, control-grid, interpolation, and quadrature errors that can be monitored separately.

5.3.1 One-step transition

Under Euler discretisation,

$$X_{n+1}^\pi = x_i + (rx_i + c_n + \beta\pi)\Delta t + \sigma\pi\sqrt{\Delta t}Z, \quad Z \sim N(0, 1), \quad (49)$$

where $c_n = c_{t_n}$. For Hermite nodes and weights $(z_k, \omega_k)_{k=1}^K$,

$$\int_{-\infty}^{\infty} e^{-z^2} f(z) dz \approx \sum_{k=1}^K \omega_k f(z_k), \quad (50)$$

and hence

$$\mathbb{E}[f(Z)] \approx \sum_{k=1}^K w_k f(\xi_k), \quad \xi_k = \sqrt{2}z_k, \quad w_k = \omega_k/\sqrt{\pi}, \quad \sum_k w_k = 1. \quad (51)$$

The one-step candidate nodes are

$$\hat{x}_{n+1}^{(k)}(x_i, \pi) = x_i + (rx_i + c_n + \beta\pi)\Delta t + \sigma\pi\sqrt{\Delta t}\xi_k. \quad (52)$$

5.3.2 Interpolation operator

Because candidate nodes generally do not coincide with wealth-grid points, linear interpolation $\mathcal{J}_{n+1}$ gives

$$\mathbb{E}[V_{n+1}(X_{n+1}^\pi) \mid X_n = x_i] \approx \sum_{k=1}^K w_k (\mathcal{J}_{n+1} V_{n+1})(\hat{x}_{n+1}^{(k)}(x_i, \pi)). \quad (53)$$

Linear interpolation is also used on the non-uniform grid. Higher-order interpolation can impair monotonicity, so the baseline implementation remains linear.

5.4 Embedded quadratic-loss HJB for PCMV and DOMV

5.4.1 Continuous-time equation

For terminal target m , solve

$$V(t, x; m) = \inf_{\pi \in \mathcal{A}_{t,x}^X} \mathbb{E}_{t,x} [(X_T + D_T - m)^2], \quad (54)$$

with

$$V(T, x; m) = (x + D_T - m)^2 \quad (55)$$

and formal HJB equation

$$V_t + \inf_{0 \leq \pi \leq x} \left\{ (rx + c_t + \beta\pi)V_x + \frac{1}{2}\sigma^2\pi^2 V_{xx} \right\} = 0. \quad (56)$$

The local Hamiltonian is

$$\mathcal{H}^Q(t, x, \pi; V) = (rx + c_t + \beta\pi)V_x + \frac{1}{2}\sigma^2\pi^2 V_{xx}. \quad (57)$$

If $V_{xx} > 0$, the local candidate and projected diagnostic are

$$\pi^{\text{loc}}(t, x) = -\frac{\beta V_x(t, x)}{\sigma^2 V_{xx}(t, x)}, \quad (58)$$

$$\pi^{\text{str}}(t, x) = \text{Proj}_{[0, x]}(\pi^{\text{loc}}(t, x)). \quad (59)$$

The numerical baseline solves the constrained recursion directly rather than relying solely on finite-difference derivatives.

5.4.2 Markov–Gauss–Hermite backward recursion

Set

$$V_{N_t}(x_i; m) = (x_i + D_T - m)^2. \quad (60)$$

For $n = N_t - 1, \dots, 0$,

$$V_n(x_i; m) = \min_{\pi \in [0, x_i]} \sum_{k=1}^K w_k(J_{n+1} V_{n+1}(\cdot; m)) (\hat{x}_{n+1}^{(k)}(x_i, \pi)), \quad n = N_t - 1, \dots, 0. \quad (61)$$

and store

$$\pi_n^{\text{QL}}(x_i; m) = \arg \min_{\pi \in [0, x_i]} \sum_{k=1}^K w_k(J_{n+1} V_{n+1}(\cdot; m)) (\hat{x}_{n+1}^{(k)}(x_i, \pi)). \quad (62)$$

This is the directly constrained feedback for the fixed-target quadratic-loss problem; the superscript QL distinguishes the quadratic-loss policy from the second-moment function Q .

Remark 7 (Semi-Lagrangian interpretation). Equation (61) follows the one-step transition distribution, interpolates the continuation value, and evaluates its conditional expectation. The constraint enters naturally through the candidate set. The same discrete kernel is subsequently used for forward propagation.

5.4.3 PCMV target consistency under the scalarised objective

For a target expected wealth, define $\mu^m(t, x) := \mathbb{E}_{\pi^{\text{QL}}(m)}^{\pi^{\text{QL}}(m)}[X_T + D_T]$ and search for m satisfying the target at $(0, x_0)$. Under the scalarised objective $\mathbb{E}[W_T] - (\gamma/2) \text{Var}(W_T)$,

$$\sup_{\pi} \left\{ \mathbb{E}[W_T] - \frac{\gamma}{2} \text{Var}(W_T) \right\} = \sup_{m \in \mathbb{R}} \left\{ m - \frac{1}{2\gamma} - \frac{\gamma}{2} V(t, x; m) \right\}, \quad (63)$$

and an interior solution satisfies

$$m^*(t, x) = \mu^*(t, x) + \frac{1}{\gamma}. \quad (64)$$

PCMV maximises (63) once at $(0, x_0)$ and commits to the resulting fixed-target feedback.

5.4.4 DOMV as rolling target selection

At each (t, x) , DOMV solves

$$\sup_{\pi \in \mathcal{A}_{t,x}^X} \left\{ \mathbb{E}_{t,x}[W_T] - \frac{\gamma_D}{2} \text{Var}_{t,x}(W_T) \right\}. \quad (65)$$

For each candidate m_ℓ in a target grid $\mathcal{M}$, the fixed-target recursion (61) is solved to completion. At node (t_n, x_i) , select

$$m_D^*(t_n, x_i) \in \arg \max_{m_\ell \in \mathcal{M}} \left\{ m_\ell - \frac{1}{2\gamma_D} - \frac{\gamma_D}{2} V_n(x_i; m_\ell) \right\}. \quad (66)$$

and implement only

$$\pi_n^{\text{DOMV}}(x_i) = \pi_n^{\text{QL}}(x_i; m_D^*(t_n, x_i)). \quad (67)$$

A candidate target is not changed midway through its backward evaluation; otherwise the calculation would no longer represent either a fixed-target problem or rolling DOMV. Mean-consistency error and avoidance of target-grid endpoints are monitored. A full independent target-grid convergence study remains future work.

5.5 Extended-HJB solver for cTCMV and dTCMV

5.5.1 Moment functions

The time-consistent MV objective is

$$J(t, x) = M(t, x) - \frac{\Gamma(t, x)}{2} (Q(t, x) - M(t, x)^2), \quad (68)$$

where

$$M(t, x) = \mathbb{E}_{t,x}^{\pi^*}[X_T + D_T], \quad (69)$$

$$Q(t, x) = \mathbb{E}_{t,x}^{\pi^*}[(X_T + D_T)^2]. \quad (70)$$

For cTCMV, $\Gamma(t, x) = \gamma_c$; for dTCMV, $\Gamma(t, x) = \rho_d/(x + H_t)$.

5.5.2 Discrete one-period equilibrium condition

Use candidate control π only from t_n to t_{n+1} and follow the already-computed equilibrium feedback thereafter. Then

$$M_n(x_i; \pi) = \sum_{k=1}^K w_k (\mathcal{J}_{n+1} M_{n+1}) (\hat{x}_{n+1}^{(k)}(x_i, \pi)), \quad (71)$$

$$Q_n(x_i; \pi) = \sum_{k=1}^K w_k (\mathcal{J}_{n+1} Q_{n+1}) (\hat{x}_{n+1}^{(k)}(x_i, \pi)). \quad (72)$$

The one-period deviation objective is

$$\Psi_n(x_i; \pi) = M_n(x_i; \pi) - \frac{\Gamma(t_n, x_i)}{2} (Q_n(x_i; \pi) - M_n(x_i; \pi)^2). \quad (73)$$

with equilibrium feedback

$$\pi_n^{\text{TC}}(x_i) \in \arg \max_{\pi \in [0, x_i]} \Psi_n(x_i; \pi). \quad (74)$$

After choosing the control,

$$M_n(x_i) = M_n(x_i; \pi_n^{\text{TC}}(x_i)), \quad (75)$$

$$Q_n(x_i) = Q_n(x_i; \pi_n^{\text{TC}}(x_i)), \quad (76)$$

$$J_n(x_i) = M_n(x_i) - \frac{\Gamma(t_n, x_i)}{2} (Q_n(x_i) - M_n(x_i)^2). \quad (77)$$

with terminal conditions

$$M_{N_t}(x_i) = x_i + D_T, \quad Q_{N_t}(x_i) = (x_i + D_T)^2, \quad J_{N_t}(x_i) = x_i + D_T. \quad (78)$$

Remark 8 (Relationship with the extended HJB). Equation (74) is the discrete one-step analogue of the continuous-time short-deviation equilibrium condition. Direct evaluation of the one-step conditional moments is comparatively robust near the low-balance boundary.

5.5.3 Equilibrium algorithm

Both cTCMV and dTCMV start from the same terminal moments and recurse backward. At each node, one-step moments are evaluated for every admissible control, the deviation objective is maximised, and the induced moments are stored. The only structural difference is

$$\Gamma_{n,i} = \gamma_c \quad \text{for cTCMV}, \quad \Gamma_{n,i} = \frac{\rho_d}{x_i + H_{t_n}} \quad \text{for dTCMV}. \quad (79)$$

Thus one solver supports both the standard-default and total-pension-wealth advisory layers.

5.5.4 CP in the common engine

CP requires no backward HJB solution. Its rule (42) is propagated with the same Gauss–Hermite transition kernel. Hence differences from the four MV strategies reflect feedback rules rather than random seeds or different distribution methods. Under equal-mean comparison, θ_{cp} is calibrated numerically.

Remark 9 (No policy iteration in the baseline). The baseline uses direct Markov–Gauss–Hermite backward recursion and direct search over $[0, x]$. It does not use Howard policy iteration for an implicit finite-difference system.

5.6 Numerical implementation of directly constrained and clipped controls

The directly constrained policy is obtained from the concept-specific backward problem already defined above: PCMV uses the fixed-target quadratic-loss minimiser after the time-zero target has been selected, DOMV uses its locally re-selected target, and cTCMV/dTCMV use the one-period equilibrium maximiser. Thus the common notation for the stored strict feedback is

$$\pi_n^{\text{str}}(x_i) = \begin{cases} \pi_n^{\text{QL}}(x_i; m_P^*), & \text{PCMV,} \\ \pi_n^{\text{DOMV}}(x_i), & \text{DOMV,} \\ \pi_n^{\text{TC}}(x_i), & \text{cTCMV/dTCMV,} \end{cases} \quad (80)$$

where the cTCMV and dTCMV cases differ through their respective $\Gamma(t_n, x_i)$. The clipped policy is

$$\pi_n^{\text{clip}}(x_i) = \text{Proj}_{[0, x_i]}(\pi^{\text{unc}}(t_n, x_i + H_{t_n})). \quad (81)$$

Each policy is propagated under its own induced wealth distribution. The comparison therefore includes both the contemporaneous control gap and the effect of that gap on future reachable states.

5.6.1 Projection-gap measures

Define

$$\Delta\pi_n(x_i) = \pi_n^{\text{str}}(x_i) - \pi_n^{\text{clip}}(x_i), \quad (82)$$

$$\Delta g_n(x_i) = \frac{\pi_n^{\text{str}}(x_i)}{x_i} - \frac{\pi_n^{\text{clip}}(x_i)}{x_i}, \quad x_i > 0. \quad (83)$$

The probability-weighted mean absolute difference is

$$\text{PG}_{\text{WMAE}} = \frac{1}{N_t} \sum_{n=0}^{N_t-1} \sum_{i=1}^I p_{n,i}^{\text{str}} |\Delta g_n(x_i)|. \quad (84)$$

Here $p_{n,i}^{\text{str}}$ is the forward distribution induced by the directly constrained policy, so the weighting evaluates the projection gap on states reached by the reference strict policy.

5.6.2 Free-boundary classification

Classify nodes by

$$\mathcal{R}_0 = \{(n, i) : \pi_n^{\text{str}}(x_i) = 0\}, \quad (85)$$

$$\mathcal{R}_I = \{(n, i) : 0 < \pi_n^{\text{str}}(x_i) < x_i\}, \quad (86)$$

$$\mathcal{R}_U = \{(n, i) : \pi_n^{\text{str}}(x_i) = x_i\}. \quad (87)$$

These are the no-risky, interior, and all-risky regions. Numerical free boundaries are the interfaces between the interior and binding regions. The constrained free boundary is kept distinct from the binding pattern of the clipped approximation.

5.7 Forward distribution propagation without random Monte Carlo

5.7.1 Probability-mass propagation on the wealth grid

After solving a feedback $\pi_n^*(x_i)$, initialise

$$p_{0,i} = \mathbf{1}_{\{x_i=x_0\}}, \quad (88)$$

with linear allocation to neighbouring nodes if necessary. From node x_i and quadrature node k ,

$$\hat{x}_{n+1}^{(k)}(x_i, \pi_n^*(x_i)) = x_i + (rx_i + c_n + \beta\pi_n^*(x_i))\Delta t + \sigma\pi_n^*(x_i)\sqrt{\Delta t}\xi_k. \quad (89)$$

If $x_\ell \leq \hat{x}_{n+1}^{(k)}(x_i, \pi_n^*(x_i)) \leq x_{\ell+1}$, allocate mass linearly:

$$p_{n+1,\ell} \leftarrow p_{n+1,\ell} + p_{n,i}w_k \frac{x_{\ell+1} - \hat{x}_{n+1}^{(k)}(x_i, \pi_n^*(x_i))}{x_{\ell+1} - x_\ell}, \quad (90)$$

$$p_{n+1,\ell+1} \leftarrow p_{n+1,\ell+1} + p_{n,i}w_k \frac{\hat{x}_{n+1}^{(k)}(x_i, \pi_n^*(x_i)) - x_\ell}{x_{\ell+1} - x_\ell}. \quad (91)$$

Transitions below $x_0 = 0$ are assigned to the lower boundary and transitions above $x_{\max}$ to the upper boundary. The pre-assignment excursion mass is recorded separately in Section 5.9. This produces a deterministic discrete law for X_T and hence for total terminal wealth $X_T + D_T$.

5.7.2 Moments and tail-risk measures

From $p_{N_t,i}$,

$$\hat{\mu}_T = \sum_i p_{N_t,i}(x_i + D_T), \quad (92)$$

$$\hat{\sigma}_T^2 = \sum_i p_{N_t,i}\{(x_i + D_T) - \hat{\mu}_T\}^2, \quad (93)$$

$$\hat{m}_{3,T} = \sum_i p_{N_t,i}\{(x_i + D_T) - \hat{\mu}_T\}^3. \quad (94)$$

The CDF is

$$F_{N_t}(x_\ell) = \sum_{i \leq \ell} p_{N_t,i}. \quad (95)$$

Let $W_i = x_i + D_T$, define $F_T^W(w) := \sum_{i: W_i \leq w} p_{N_t, i}$, and set $q_\tau := \inf\{w : F_T^W(w) \geq \tau\}$. Because the propagated law is discrete, the probability mass at the quantile boundary is included only as needed to make the tail probability exactly α . The lower-tail conditional mean is

$$\text{LCVaR}_\alpha = \frac{1}{\alpha} \left[\sum_{W_i < q_\alpha} p_{N_t, i} W_i + \left(\alpha - \sum_{W_i < q_\alpha} p_{N_t, i} \right) q_\alpha \right]. \quad (96)$$

and the mean of the upper α tail is

$$\text{UCVaR}_\alpha = \frac{1}{\alpha} \left[\sum_{W_i > q_{1-\alpha}} p_{N_t, i} W_i + \left(\alpha - \sum_{W_i > q_{1-\alpha}} p_{N_t, i} \right) q_{1-\alpha} \right]. \quad (97)$$

5.7.3 Summarising path-dependent glide paths

For $n \geq 1$, the distribution-weighted mean glide path is

$$\bar{g}_n = \sum_{i: x_i > 0} p_{n, i} g_n(x_i). \quad (98)$$

At the deterministic initial balance, interpolating ratios would give undue influence to the arbitrary $g(0, 0) = 0$. We therefore interpolate the amount,

$$\pi_0 = \text{Interp}_x \{x_i g_0(x_i)\} (x_0), \quad \bar{g}_0 = \frac{\pi_0}{x_0}. \quad (99)$$

and propagate the first transition from the exact x_0 . Controls are defined only on intervals $[t_n, t_{n+1})$, so glide-path figures end at $T - \Delta t$ rather than appending an artificial zero at maturity. Weighted quantiles of $g_n(x_i)$ under $p_{n, i}$ provide a deterministic counterpart to pathwise glide-path quantiles.

5.8 Rolling conditional evaluation

5.8.1 Definition of the rolling conditional outcome profile

Let $W_T^{j; t, x}$ denote terminal wealth when the stored feedback for strategy j is continued from date t and balance x . Define

$$\begin{aligned} \mathbf{O}_j(t, x; \alpha) = & (\mu_j(t, x), s_j(t, x), q_{j, \alpha}(t, x), q_{j, 0.50}(t, x), q_{j, 1-\alpha}(t, x), \\ & \text{LCVaR}_{j, \alpha}(t, x), \text{UCVaR}_{j, \alpha}(t, x), \text{Skew}_j(t, x), g_j(t, x), \pi_j(t, x)). \end{aligned} \quad (100)$$

This is a multi-dimensional profile rather than a scalar ranking: it reports conditional mean, dispersion, quantiles, lower and upper tails, skewness, and the current risky share and amount.

5.8.2 Law of total expectation and variance

For strategy j , let X_t^j be the account balance and $W_T^j = X_T^j + D_T$. Define

$$\mu_j(t, x) = \mathbb{E}[W_T^j \mid X_t^j = x], \quad v_j(t, x) = \text{Var}(W_T^j \mid X_t^j = x). \quad (101)$$

Proposition 1 (Decomposition of time-zero and rolling evaluation). *If the required second moments are finite, then for every $t \in [0, T]$,*

$$\mathbb{E}[W_T^j] = \mathbb{E}[\mu_j(t, X_t^j)], \quad (102)$$

$$\text{Var}(W_T^j) = \mathbb{E}[v_j(t, X_t^j)] + \text{Var}(\mu_j(t, X_t^j)). \quad (103)$$

Proof. The first identity is iterated expectation. For the second, decompose terminal wealth into the innovation around its conditional mean and the deviation of that conditional mean from the unconditional mean; the cross term has zero conditional expectation. $\square$

The first variance term is the average future process risk remaining after date t . The second is cross-state dispersion in expected terminal wealth generated by investment experience up to t . At time zero, the initial state is deterministic and the second term is zero. At later dates, unconditional variance mixes future risk with heterogeneity already created. Conditioning on the observed balance fixes that historical state difference and isolates the distribution relevant to prospective decisions.

This distinction is especially important in comparing PCMV and dTCMV. PCMV is designed to minimise total terminal variance at time zero under commitment, so its concentration in a time-zero comparison is natural. dTCMV changes expected outcomes, downside risk, and upside participation according to the current balance and background wealth. Its value and cost therefore require state-fixed rolling evaluation as well as

unconditional comparison. This does not claim that dTCMV dominates PCMV; the metrics answer different decision-date questions.

We use two visualisations. The own-path profile starts from low, median, and high wealth quantiles generated by each strategy,

$$\mathbf{O}_j(t_n, q_{n,\tau}^j; \alpha), \quad \tau \in \{0.10, 0.50, 0.90\}. \quad (104)$$

The common-state profile starts multiple strategies from the same (t, x) and isolates the effect of changing only the future feedback rule.

5.8.3 Conditional starting states and remaining-horizon laws

For strategy j , representative own-path starting states are

$$q_{n,0.10}^j, \quad q_{n,0.50}^j, \quad q_{n,0.90}^j. \quad (105)$$

Starting from (t_n, x_i) , the stored feedback is propagated to maturity. The conditional terminal CDF is

$$F_j(z | t_n, x_i) := \Pr(X_T^j + D_T \leq z | X_{t_n}^j = x_i). \quad (106)$$

Conditional moments, quantiles, LCVaR, UCVaR, and skewness are computed from this law. To isolate personalisation from different prior wealth paths, cTCMV and dTCMV are also compared from the same dTCMV-generated state.

5.8.4 Why rolling evaluation is needed

Time-zero terminal distributions remain essential for initial plan design and naturally favour PCMV's variance compression at a matched mean. However, after 10, 20, or 30 years, the current balance is observed and past outcomes are given. The relevant question is the distribution remaining from that state, not a re-counting of past cross-state dispersion as future risk. Reporting both unconditional and rolling conditional profiles therefore reconciles PCMV's time-zero efficiency with dTCMV's state-specific personalisation.

5.9 Numerical diagnostics

5.9.1 Moment consistency

For cTCMV and dTCMV, backward moments must agree with moments of the forward terminal distribution:

$$\varepsilon_M = \left| M_0(x_0) - \sum_i p_{N_t, i}(x_i + D_T) \right|, \quad (107)$$

$$\varepsilon_Q = \left| Q_0(x_0) - \sum_i p_{N_t, i}(x_i + D_T)^2 \right|. \quad (108)$$

5.9.2 Mass conservation and boundary aggregation

Let $\tilde{p}_{n+1, \ell}$ be the unnormalised next-period mass and

$$P_{n+1}^{\text{tot}} = \sum_{\ell} \tilde{p}_{n+1, \ell}. \quad (109)$$

Record

$$\varepsilon_p(n+1) = |1 - P_{n+1}^{\text{tot}}|. \quad (110)$$

before normalisation. Upper and lower excursions before boundary assignment are

$$e_n^{\text{up}} = \sum_{i,k} p_{n,i} w_k \mathbf{1}_{\{\hat{x}_{n+1}^{(k)}(x_i, \pi_n^*(x_i)) > x_{\max}\}}, \quad (111)$$

$$e_n^{\text{low}} = \sum_{i,k} p_{n,i} w_k \mathbf{1}_{\{\hat{x}_{n+1}^{(k)}(x_i, \pi_n^*(x_i)) < 0\}}. \quad (112)$$

We distinguish maximum excursion by date, cumulative flow, and terminal boundary mass. Since the same mass can hit the boundary repeatedly, cumulative flow is not itself an exceedance probability. Upper-bound effects are assessed by nested domains that extend the tail while retaining the baseline interval $[0, 300]$.

6 Numerical Results

Main findings. First, the clipped approximation reproduces directly constrained cTCMV well under the baseline calibration. Second, for DOMV and dTCMV, clipping does not adequately reproduce either policy shape or the lower tail, so internalising the constraint has material value. Third, dTCMV generates an endogenous U-shaped glide path through the interaction of total pension wealth, remaining horizon, and constrained continuation value. Fourth, rolling conditional profiles reveal state-specific benefits and costs hidden behind a common time-zero mean, particularly the way dTCMV exchanges additional downside risk for greater upside participation.

6.1 Baseline case: $D_T = 0$

Set $D_T = 0$. Taking the PCMV mean as the common target, calibrate the DOMV, cTCMV, and dTCMV variance-aversion parameters and the CP risky share. All strategies have expected terminal wealth of 84.78 to two decimal places, so the comparison focuses on variance, lower and upper tails, and the timing of risk-taking.

6.2 Comparison design

The analysis is divided into two stages. The principal cross-concept comparison evaluates the *directly constrained* PCMV, DOMV, cTCMV, and dTCMV policies and CP under common market and plan assumptions and a common expected terminal wealth. Section 6.7 separately evaluates approximation accuracy by comparing directly constrained feedback with the corresponding clipped control. The unconstrained policies themselves are not included in terminal-distribution rankings because they may violate the DC constraint.

The monthly PCMV mean

$$\bar{m} = 84.777086$$

is used as the common target. For $j \in \mathcal{J} \cup \{\text{CP}\}$, parameters are selected so that

$$|\mathbb{E}[W_T^j] - \bar{m}| \rightarrow 0. \quad (113)$$

The maximum mean error in the monthly calculation is below 0.003.

For PCMV, DOMV, and cTCMV the objective is written

$$\mathbb{E}[W_T] - \frac{\gamma_j}{2} \text{Var}(W_T), \quad j \in \{p, D, c\}, \quad (114)$$

where $W_T = X_T + D_T$. dTCMV uses $\Gamma(t, x) = \rho_d/(x + H_t)$. The numerical magnitudes of γ_j and ρ_d are not directly comparable because they have different roles and scales. Calibration values and achieved means are recorded in the reproducibility data.

PCMV selects the time-zero target m_P^* once from the envelope of fixed-target quadratic-loss problems. DOMV compares the same family at every date and state, executes the first control from the selected problem, and repeats at the next date. cTCMV and dTCMV maximise the one-step deviation objective (74). All four therefore impose $0 \leq \pi \leq x$ inside the backward computation.

6.3 Numerical specification

The baseline uses monthly decisions for 40 years,

$$N_t = 480, \quad \Delta t = 1/12,$$

with $D_T = 0$ and hence $W_T = X_T$. Other assumptions appear in Table 6.

Table 6: Common numerical parameters for the monthly equal-mean baseline with $D_T = 0$

Item	Symbol	Value
Accumulation horizon	T	40 years
Monthly decision grid	N_t	480
Initial DC balance	x_0	1/12
Contribution flow	c	1 per year
Terminal DB benefit	D_T	0
Risk-free rate / risky expected return	(r, μ)	(0.015, 0.055)
Volatility	σ	0.180
Equal-mean CP risky share	θ_{cp}	0.4695
Wealth grid	$(x_{\max}, N_x)$	(300, 151), non-uniform
Control candidates / GH order	(N_a, K)	(15, 5), with local refinement
DOMV target grid	$\mathcal{M}$	40, 45, ..., 340

Terminal distributions are obtained by forward propagation with the same Gauss–Hermite transitions used in the backward solution. Directional sensitivity to D_T , r , μ , σ , and the contribution profile is provided on the English supplementary-figures page of the public repository. These are 80-step screening analyses for direction and robustness, not convergence studies.

6.4 Monthly baseline results

Table 7 reports the monthly equal-mean results. The endogenous PCMV target is $m_{P,\text{base}}^* = 104.772$. Rounded means are shown as 84.78; unrounded values and calibration errors are retained in the CSV files.

Table 7: Monthly terminal DC wealth and endogenous risky allocation under $D_T = 0$ and equal means ($N_t = 480$)

Strategy	Mean	SD	Skewness	q05	q50	q95	LCVaR _{5%}	Mean glide path	Upper-bound mass
PCMV	84.78	22.959	-1.2506	33.345	92.706	109.728	24.524	0.674	0.445
DOMV	84.78	28.625	0.4144	40.637	82.622	134.847	33.336	0.632	0.278
cTCMV	84.78	28.553	0.4143	42.160	82.622	134.847	33.385	0.624	0.259
dTCMV	84.78	41.349	1.3471	34.760	76.783	164.405	28.398	0.436	0.057
CP (equal mean)	84.78	37.927	1.3940	39.136	76.783	156.811	32.503	0.469	0.000

Notes: q05, q50, and q95 are left-continuous quantiles on the non-uniform wealth grid. “Mean glide path” is the time average of the probability-weighted share $\bar{g}_n = \sum_i p_n^i g_{n,i}$. “Upper-bound mass” is the time average of $\sum_i p_n^i \mathbf{1}_{\{g_{n,i}=1\}}$. DOMV and cTCMV share displayed q50 and q95 values because distinct CDFs cross those probability levels at the same grid nodes; their full distributions are not identical. Unrounded skewness is 0.414393 for DOMV and 0.414260 for cTCMV, and their standard deviations, q05, LCVaR, and binding masses also differ.

Equalising the mean makes the distributional differences clear. PCMV has the smallest standard deviation and the highest median, but also negative skewness and low q05 and LCVaR. Compression around the target is paid for in the left tail. DOMV and cTCMV have similar standard deviations and skewness, but cTCMV has slightly higher q05 and LCVaR, giving a relatively balanced standard-default candidate. dTCMV is the opposite of PCMV: it has the largest dispersion and positive skewness, and the largest q95. It strengthens right-tail participation without increasing the mean. Equal-mean CP also has positive skewness but lacks MV optimality and personalisation based on total pension wealth.

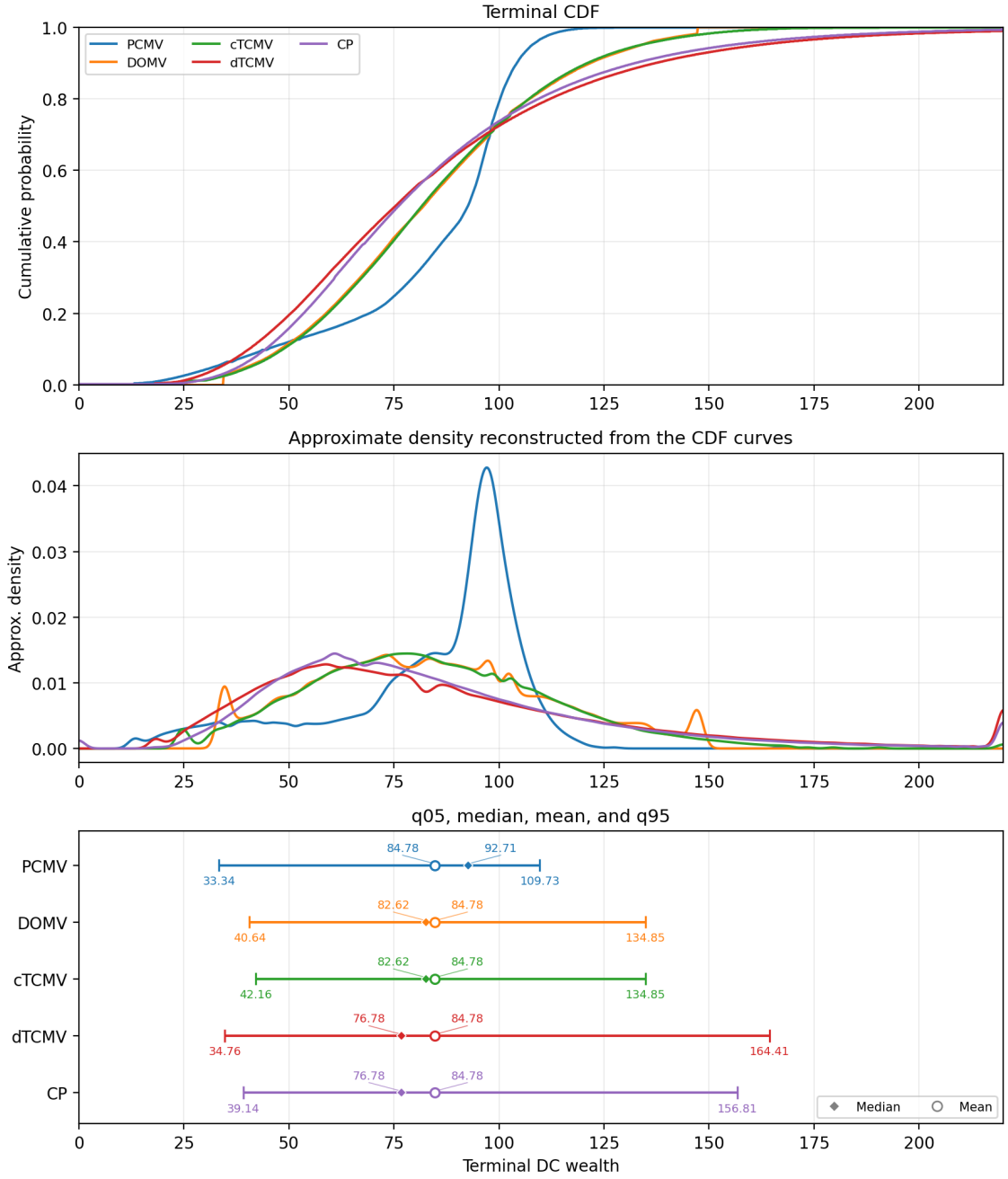


Figure 1: Monthly terminal DC wealth under $D_T = 0$ and equal-mean calibration: CDFs, approximate densities reconstructed from the plotted CDF curves, and q05–q95 interval summaries. Numerical values are printed at the corresponding q05, median, mean, and q95 locations.

Because equal-mean CP has a long right tail, its sensitivity to the finite wealth domain must be checked separately. The common-grid results report principal quantiles and means; detailed tail diagnostics are provided in the reproducibility package.

6.5 Endogenous glide paths, the upper constraint, and free boundaries

Figure 2 shows the probability-weighted risky share under the wealth distribution generated by each strategy. CP is constant by definition at approximately 46.9%. The other lines are summaries of state-dependent feedback controls, not age-only rules.

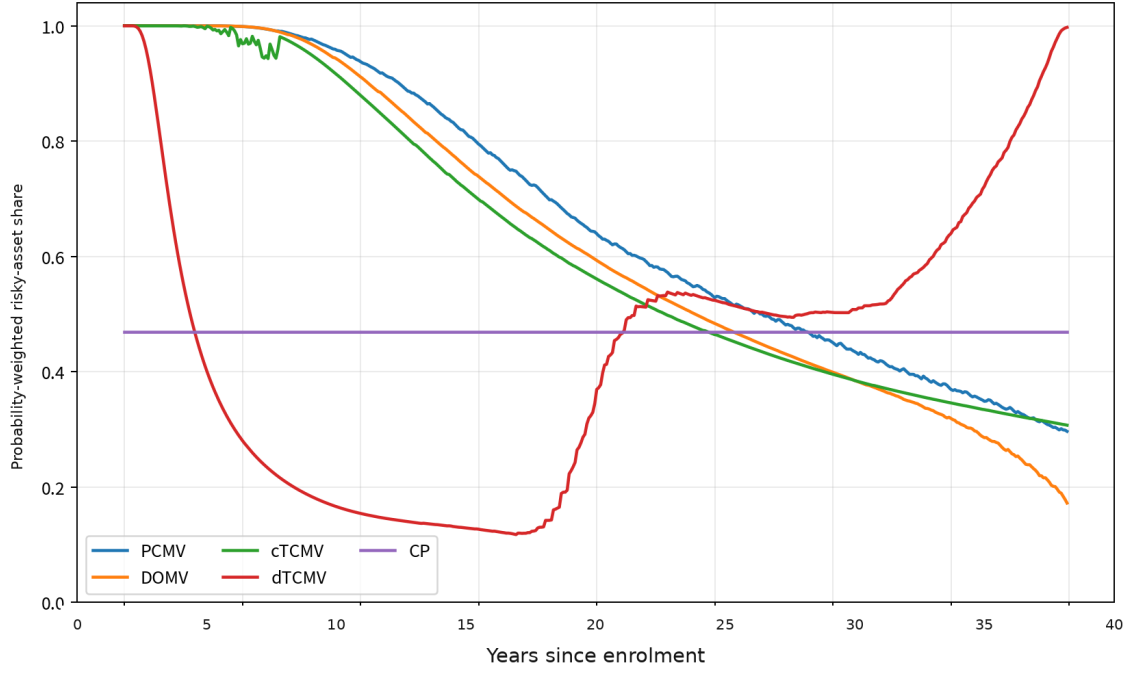


Figure 2: Probability-weighted endogenous glide paths on the decision grid $0 \leq t < T$, with the equal-mean CP benchmark

Using the investment-amount interpolation in (99), PCMV, DOMV, cTCMV, and dTCMV all begin at $\bar{g}_0 = 1$, whereas CP begins at 0.469. This means the upper constraint $\pi = x_0$ binds at the small positive initial balance. At the final decision date $T - \Delta t$, the mean risky shares remain positive: 0.296 for PCMV, 0.323 for DOMV, 0.331 for cTCMV, 0.474 for dTCMV, and 0.469 for CP.

The probability-weighted binding masses are 0.445, 0.278, 0.259, and 0.057 for PCMV, DOMV, cTCMV, and dTCMV respectively. This measure indicates how frequently members are likely to experience the upper investment constraint.

Mechanism behind the dTCMV U-shape. The late increase is not explained by declining background wealth alone. In the unconstrained representation, the risky share can be written $\theta_{\rho_d}(t)(1 + H_t/x)$. The factor H_t/x declines with age and therefore pushes the share down, while the unconstrained equilibrium coefficient $\theta_{\rho_d}(t)$ rises towards retirement as the solution of the integral equation. For the constrained policy, define the descriptive effective coefficient

$$\hat{\theta}_t^{\text{str}} := \frac{g^{d,\text{str}}(t, x_t^{50})}{1 + H_t/x_t^{50}},$$

where x_t^{50} is the median dTCMV wealth state. This is not the unconstrained θ_{ρ_d} ; it incorporates the constrained continuation value.

Table 8: Decomposition of the dTCMV U-shaped glide path at the median state

Elapsed years	Median x_t^{50}	H_t/x_t^{50}	$\Gamma(t, x_t^{50})$	$\theta_{\rho_d}(t)$	Direct share $g^{d,\text{str}}$	$\hat{\theta}_t^{\text{str}}$
0	0.083	360.786	0.0396	0.0890	1.000	0.0028
10	11.000	2.195	0.0340	0.1819	0.143	0.0447
20	24.074	0.717	0.0289	0.3269	0.364	0.2119
30	45.268	0.205	0.0219	0.5729	0.490	0.4066
35	58.507	0.082	0.0188	0.7642	0.643	0.5940
39	72.980	0.014	0.0161	0.9719	0.929	0.9161

Initially, background wealth is very large relative to the investable DC balance, so the unconstrained share far exceeds one and the upper constraint caps it. As H_t/x_t^{50} falls and the policy leaves the binding region, the risky share declines. Meanwhile, $\theta_{\rho_d}(t)$ rises from 0.0890 to 0.9719. Until the middle years, the decline in $1 + H_t/x_t^{50}$ dominates; later, the rising equilibrium coefficient dominates. The effective constrained coefficient moves in the same direction, with the constrained continuation value adjusting its magnitude. The U-shape is therefore generated by the interaction of background wealth relative to the account, the intrinsic time dependence of the unconstrained equilibrium, and the upper investment limit.

A binding set need not be one connected interval. A practically irrelevant high-wealth binding region may coexist with the main low-wealth region. We therefore emphasise probability-weighted binding and reachable-region policy surfaces rather than reporting only the maximum wealth at which a constraint binds.

6.6 cTCMV and dTCMV feedback surfaces and projection gaps

For this diagnostic, coefficient normalisations are selected to make approximation errors comparable; the exercise is not the same as the equal-mean distribution comparison in Table 7.

Figure 3 restricts the policy surfaces to the 0.5th–99.5th percentile envelope of each strategy’s forward distribution. cTCMV binds at the upper constraint for low balances and moves towards an approximately constant-dollar interior control. dTCMV leaves the binding region and forms a total-pension-wealth-dependent, state-proportional interior control.

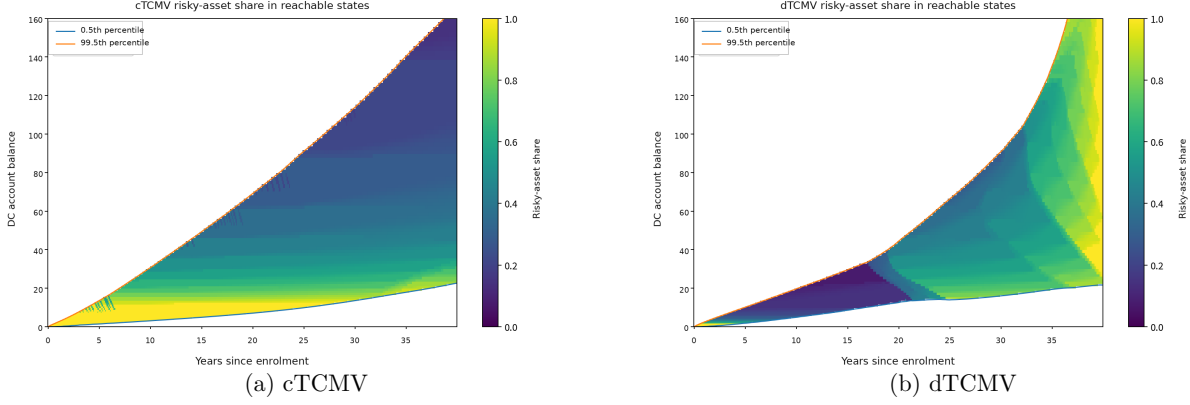


Figure 3: Directly constrained feedback surfaces within the 99% reachable envelope

Table 9 reports reachable-probability-weighted differences from the clipped unconstrained policies on a refined grid $(N_x, N_a, K) = (1001, 81, 15)$. Average share differences are small, but local maximum differences reach approximately 0.20 for cTCMV and 0.14 for dTCMV. Clipping may therefore be a useful average approximation while failing to reproduce state-level advice near a free boundary.

Table 9: Projection-gap diagnostics on the monthly diagnostic grid

Strategy	PG_{WMAE}	$ \Delta g $ q95	$ \Delta g $ q99	Max $ \Delta g $	Weighted mean $ \Delta \pi $	Max $ \Delta \pi $
cTCMV	0.0210	0.1120	0.1480	0.2014	0.506	3.771
dTCMV	0.0363	0.0686	0.0834	0.1447	0.947	4.970

6.7 Directly constrained feedback versus clipped approximation

For each $j \in \mathcal{J}$, compare probability-weighted glide paths under the separate forward distributions generated by the directly constrained and clipped policies:

$$\bar{g}_j^{\text{str}}(t_n) = \sum_i p_{n,i}^{j,\text{str}} a_{n,i}^{j,\text{str}}, \quad \bar{g}_j^{\text{clip}}(t_n) = \sum_i p_{n,i}^{j,\text{clip}} a_{n,i}^{j,\text{clip}}.$$

The time-average absolute glide-path difference is

$$G_{MAE}^j = \frac{1}{N_t} \sum_{n=0}^{N_t-1} |\bar{g}_j^{\text{str}}(t_n) - \bar{g}_j^{\text{clip}}(t_n)|. \quad (115)$$

This differs from the node-level metric (84).

The PCMV diagnostic uses $m_{P,\text{diag}}^* = 109.157$, distinct from the equal-mean baseline target $m_{P,\text{base}}^* = 104.772$. DOMV uses $\lambda_D = \gamma_D/2 = 0.025$ in (13); cTCMV and dTCMV use (14) and (15).

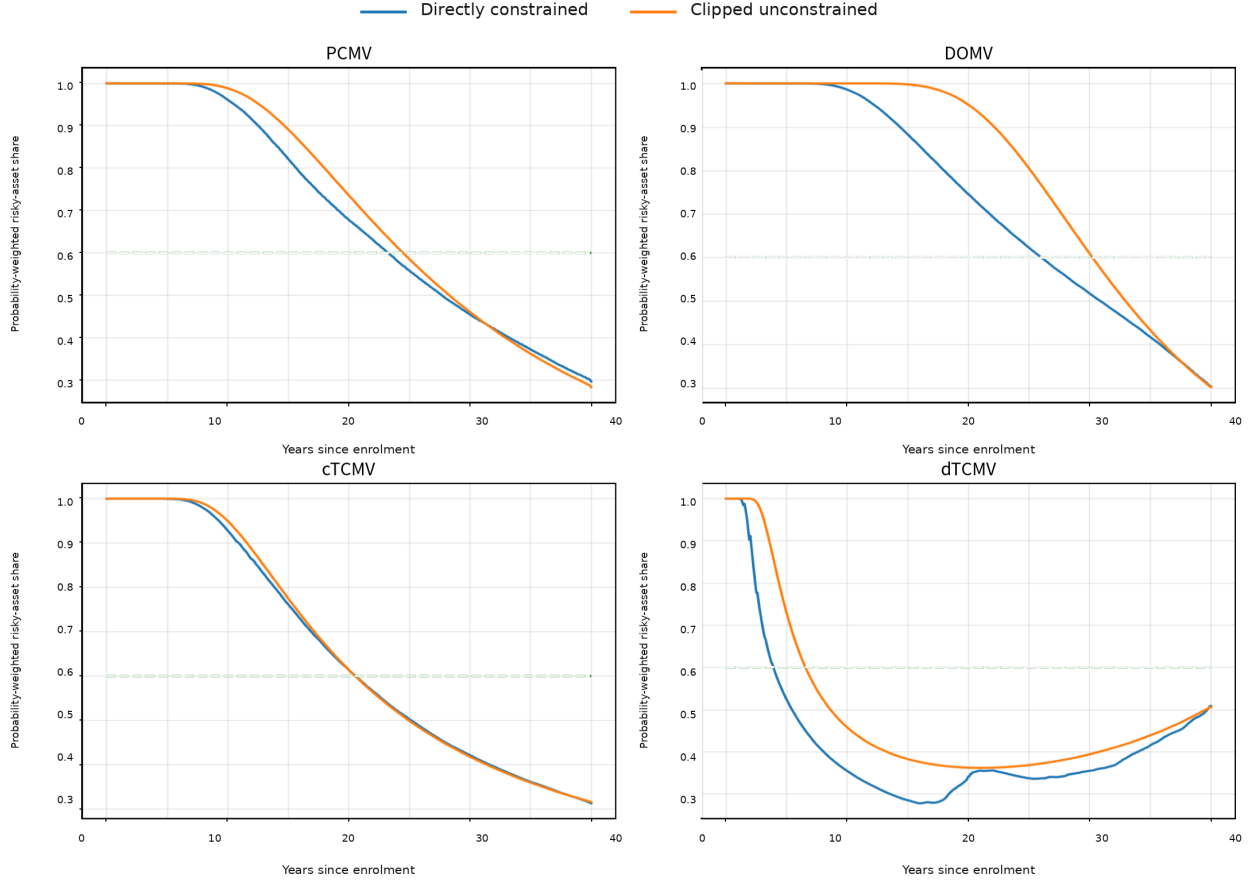


Figure 4: Probability-weighted glide paths: directly constrained feedback versus clipped approximation, by strategy

cTCMV remains close to its clipped approximation throughout the horizon. DOMV clipping stays substantially more aggressive through the middle years. dTCMV clipping is also more aggressive in the early and middle years and returns more smoothly after its trough. PCMV gaps are smaller but still relevant to the lower tail.

Table 10: Summary of directly constrained versus clipped differences under the diagnostic setting of Figure 4

Strategy	G_{MAE}	$\text{Max } \Delta \bar{g} $	$\Delta \mathbb{E}[W_T]$	$\Delta q_{0.05}$	$\Delta \text{LCVaR}_{5\%}$
PCMV	0.025	0.074	0.869	-1.609	-1.033
DOMV	0.078	0.211	7.434	-4.496	-4.475
cTCMV	0.006	0.025	-0.010	0.587	-0.278
dTCMV	0.069	0.272	3.440	-0.590	-0.627

Δ denotes clipped minus directly constrained. Each outcome measure is calculated under the terminal distribution generated by its own policy.

For cTCMV, the mean glide-path gap is 0.006 and the maximum is 0.025, with little effect on expected wealth or the lower 5% point. The clipped closed-form solution is therefore a strong practical approximation under this calibration. For PCMV, the average gap is 0.025; clipping slightly raises the mean but worsens q_{05} and LCVaR, so lower-tail applications require separate checking. DOMV has an average gap of 0.078 and a maximum of 0.211. Clipping raises the mean but materially worsens the lower tail, indicating that rolling re-optimisation is not reproduced by a static clipped rule. dTCMV has an average gap of 0.069 and a maximum of 0.272; clipping produces a more aggressive allocation, a higher mean, and slightly weaker lower-tail outcomes. Approximation accuracy is therefore strongly solution-concept dependent.

6.8 CP benchmark under equal-mean design

At the common mean, dTCMV has a larger q_{95} but slightly lower q_{05} and LCVaR than CP, while CP has lower standard deviation. This is consistent with the possibility of partial first-order dominance by a constant-proportion strategy in a low-wealth region reported by [van Staden et al. \(2021\)](#), but the two tail statistics alone do not establish that dominance for the constrained distributions here. Since dTCMV has the larger right tail, neither strategy dominates over the full distribution. The distinctive role of dTCMV is state-dependent adjustment using balance and background wealth, not uniform superiority to CP at the group level.

6.9 Rolling conditional profiles and numerical validation

Table 11 starts each strategy from the median of its own intermediate wealth distribution and propagates the same stored feedback to maturity. This is an own-path visualisation. PCMV’s conditional standard deviation and risky share decline as maturity approaches and the fixed target becomes closer. DOMV and cTCMV also de-risk progressively at the median state. dTCMV, by contrast, falls sharply by year 10 and then rises again, reaching approximately 0.93 at year 39, while retaining larger conditional dispersion.

Table 11: Rolling conditional outcomes from representative median states under equal-mean $D_T = 0$ calibration

Strategy	Elapsed years	Median X_t	Conditional mean	Conditional SD	Risky share
PCMV	0	0.083	84.777	22.959	1.000
PCMV	20	31.952	86.151	19.528	0.683
PCMV	35	76.783	91.652	8.912	0.286
PCMV	39	90.654	93.561	3.625	0.167
DOMV	0	0.083	84.777	28.625	1.000
DOMV	20	30.582	82.253	21.183	0.568
DOMV	35	69.250	83.452	9.490	0.275
DOMV	39	80.658	83.473	3.851	0.255
cTCMV	0	0.083	84.779	28.553	1.000
cTCMV	20	30.582	82.911	21.859	0.522
cTCMV	35	67.412	82.545	10.876	0.326
cTCMV	39	78.711	81.840	4.823	0.288
dTCMV	0	0.083	84.778	41.349	1.000
dTCMV	20	24.074	82.641	38.689	0.364
dTCMV	35	58.507	79.927	27.071	0.643
dTCMV	39	72.980	78.066	14.049	0.929

Backward conditional moments agree with forward propagated moments. Across six review dates, maximum absolute errors remain below 3.5×10^{-12} for first moments and 3.6×10^{-10} for second moments.

Table 12: Backward–forward consistency of rolling conditional moments

Strategy	Max ε_M	Max ε_Q	Max SD difference
PCMV	3.41×10^{-12}	3.13×10^{-10}	5.78×10^{-12}
DOMV	3.33×10^{-12}	3.13×10^{-10}	4.39×10^{-12}
cTCMV	3.35×10^{-12}	3.16×10^{-10}	4.43×10^{-12}
dTCMV	3.37×10^{-12}	3.53×10^{-10}	2.64×10^{-12}

Probability mass and upper-bound diagnostics. Pre-normalisation mass errors are below 1.6×10^{-15} for all strategies. Lower-bound excursion mass is numerically zero. Upper-bound excursion is negligible for PCMV, DOMV, and cTCMV, but on the baseline $x_{\max} = 300$ grid dTCMV has maximum one-step excursion mass 1.09×10^{-3} and cumulative flow 1.88×10^{-2} .

Table 13: Mass conservation and upper-bound diagnostics for the $D_T = 0$ baseline

Strategy	Max ε_p	Max ω_n^+	$\sum_n \omega_n^+$	Terminal upper-bound mass
PCMV	9.99×10^{-16}	3.03×10^{-26}	1.42×10^{-25}	3.83×10^{-26}
DOMV	1.44×10^{-15}	5.62×10^{-6}	7.73×10^{-4}	1.13×10^{-8}
cTCMV	1.55×10^{-15}	4.57×10^{-11}	4.74×10^{-9}	3.16×10^{-12}
dTCMV	1.44×10^{-15}	1.09×10^{-3}	1.88×10^{-2}	1.15×10^{-3}

dTCMV upper-domain sensitivity. Keeping the baseline $[0, 300]$ grid unchanged and appending tail nodes above 300 shows that $x_{\max} = 600$ and 2100 differ by less than 0.001 in the mean, 0.006 in standard deviation, and 0.011 in $\text{UCVaR}_{5\%}$. At $x_{\max} = 900$, maximum one-step excursion mass falls to 3.45×10^{-8} . The baseline domain slightly raises the mean and understates the right tail. Recalibrating ρ_d to 1.187334 on $x_{\max} = 900$ restores the common mean and preserves the ranking that dTCMV has the greatest dispersion and right-tail participation.

Table 14: dTCMV upper-domain sensitivity and equal-mean recalibration on the wider domain

$x_{\max}$	ρ_d	Mean	SD	UCVaR _{5%}	Mean share	Max ω_n^+
300 (baseline)	1.193359	84.778	41.349	199.244	0.4357	1.09×10^{-3}
600	1.193359	84.697	42.010	202.177	0.4315	3.26×10^{-6}
900	1.193359	84.698	42.016	202.188	0.4315	3.45×10^{-8}
900 (recalibrated)	1.187334	84.777	42.168	202.766	0.4322	3.74×10^{-8}

6.9.1 dTCMV rolling evaluation by wealth quantile

Median-state evaluation alone cannot reveal how feedback responds to member wealth differences. We therefore use the 10th, 50th, and 90th percentiles of dTCMV’s own intermediate distribution,

$$x_{t,L} = q_{t,0.10}^d, \quad x_{t,M} = q_{t,0.50}^d, \quad x_{t,H} = q_{t,0.90}^d,$$

at years 10, 20, 30, 35, and 39. Conditional means, standard deviations, skewness, q05, LCVaR_{5%}, q95, and UCVaR_{5%} are recomputed from each state. Both the risky *share* and the risky *amount* $\pi^d(t, x) = xg^d(t, x)$ are needed: average age-based shares alone materially understate the policy’s state dependence. Full quantile results are stored in the accompanying CSV.

6.9.2 Common-state conditional quantiles: cTCMV versus dTCMV

To separate prior wealth-path differences from the effect of the future strategy, cTCMV and dTCMV are propagated from the same date and the same DC balance. Table 15 selects three representative states: median and high balances at year 30, and a low balance at year 35.

Table 15: Conditional outcomes from common states: cTCMV versus dTCMV under equal-mean calibration

State	Strategy	Mean	SD	Median	q05	LCVaR _{5%}	q95	Skewness
Year 30, median balance ($x = 45.27$)	cTCMV	72.41	14.80	72.98	48.46	43.47	96.86	0.18
	dTCMV	81.37	32.69	74.87	39.14	33.95	142.03	1.22
Year 30, high balance ($x = 67.41$)	cTCMV	98.82	16.29	98.96	72.98	66.63	125.49	0.13
	dTCMV	114.15	44.59	107.54	56.78	48.36	198.84	1.06
Year 35, low balance ($x = 36.20$)	cTCMV	48.29	9.28	48.46	33.34	30.04	63.79	0.16
	dTCMV	51.34	17.08	48.46	27.91	25.37	82.62	1.02

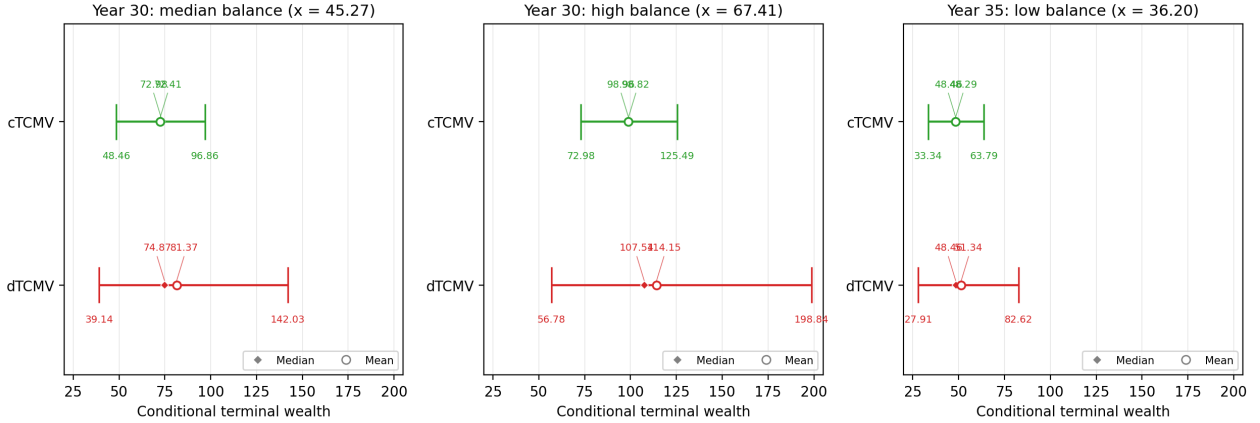


Figure 5: Common-state rolling comparison of cTCMV and dTCMV: q05–q95 intervals, conditional medians, and conditional means. Numerical values are printed at their corresponding locations.

At the year-30 median balance, dTCMV raises the conditional mean from 72.41 to 81.37 and q95 from 96.86 to 142.03, while skewness rises from 0.18 to 1.22. The cost is a rise in standard deviation from 14.80 to 32.69 and lower q05 and LCVaR. At the high year-30 balance, the asymmetry is even stronger: q95 rises from 125.49 to 198.84 and the mean from 98.82 to 114.15, but dispersion expands and lower-tail measures deteriorate. Even at the low year-35 state, dTCMV does not automatically provide downside protection under the equal-mean calibration. It raises the mean and q95 but also raises dispersion and reduces q05 and LCVaR.

The value of dTCMV is therefore not unconditional stochastic dominance. It is the capacity to change right-tail participation and risk materially according to current balance and background wealth. The common-state profile provides a governance tool for offering dTCMV as optional personalised advice rather than as the universal default. Suitability should consider appetite for upside participation, downside-risk limits, background wealth, and remaining horizon, potentially supplemented by risky-share caps or LCVaR constraints.

7 Extension to Mean–Variance–Skewness

7.1 Criterion, nesting within MV, and the position of cTCMVS

To incorporate upside asymmetry in terminal wealth explicitly, consider

$$J^{\text{MVS}} = \mathbb{E}[W_T] - \frac{\gamma}{2} \text{Var}(W_T) + \frac{\eta}{3} \mathbb{E}[(W_T - \mathbb{E}[W_T])^3]. \quad (116)$$

PCMV–MVS commits at time zero to the mean, variance, and third central moment; DOMV–MVS resolves the remaining-horizon problem at each state; and dTCMV–MVS treats total-pension-wealth-dependent variance aversion and skewness preference as an equilibrium among successive selves. MVS is therefore not a mechanical addition of the same objective to four otherwise identical models. It is a higher-moment extension that preserves the temporal decision principle of the underlying solution concept.

Proposition 2 (Reduction to MV when skewness preference is zero). *Under the same market model, constraints, variance-aversion parameter, initial condition, and solution concept, setting $\eta = 0$ makes (116) identical to the corresponding MV criterion. The sets of optimal or equilibrium controls therefore coincide. Under uniqueness, or under the same tie-breaking rule, the feedback control, glide path, and terminal distribution also coincide.*

In an independently implemented audit on the small grid $(N_t, N_x, N_a, K) = (8, 31, 7, 3)$, the maximum policy difference between MV and MVS at $\eta = 0$ was zero to machine precision for PCMV, DOMV, cTCMV, and dTCMV. This is a regression test showing that the MVS code reduces correctly to MV; it does not establish the economic usefulness of a positive skewness coefficient.

Applying the closed-form result of [Mu et al. \(2019\)](#) for the constant-coefficient unconstrained Black–Scholes problem to the constant-variance-aversion specification used for the first layer shows that the unconstrained cTCMVS equilibrium risky amount coincides with that of cTCMV. The clipped approximations constructed by projecting this common unconstrained solution onto the same interval $[0, x]$ therefore also coincide. Within the scope of the unconstrained and clipped problems, adding positive skewness preference does not produce a distinct cTCMVS glide path.

This conclusion does not automatically extend to the directly constrained problem. Once $0 \leq \pi \leq x$ is internalised in the continuation value or equilibrium moments, the probability of future binding and the conditional higher moments depend on current wealth, and the cancellation structure of the unconstrained problem need not survive. Equality between directly constrained cTCMVS and directly constrained cTCMV is therefore neither guaranteed nor proved here. Deriving the extended HJB system, sufficient or necessary equality conditions, and the associated free boundary for directly constrained cTCMVS with positive skewness preference is left for future research.

7.2 Interpretation by solution concept and coefficient sensitivity of dTCMV–MVS

The meaning of an MVS extension differs materially across solution concepts. For cTCMVS, the equality above holds for the unconstrained and clipped cases, whereas the directly constrained equality remains unresolved. By contrast, in dTCMV–MVS, where variance aversion and skewness preference depend on total pension wealth, the coefficients can alter not only the level of risky investment but also the *timing* of risk-taking. Because this timing is central to the endogenous glide-path question, the coefficient comparison reported as Supplementary Figure 5.1 in the public repository is included in the main paper.

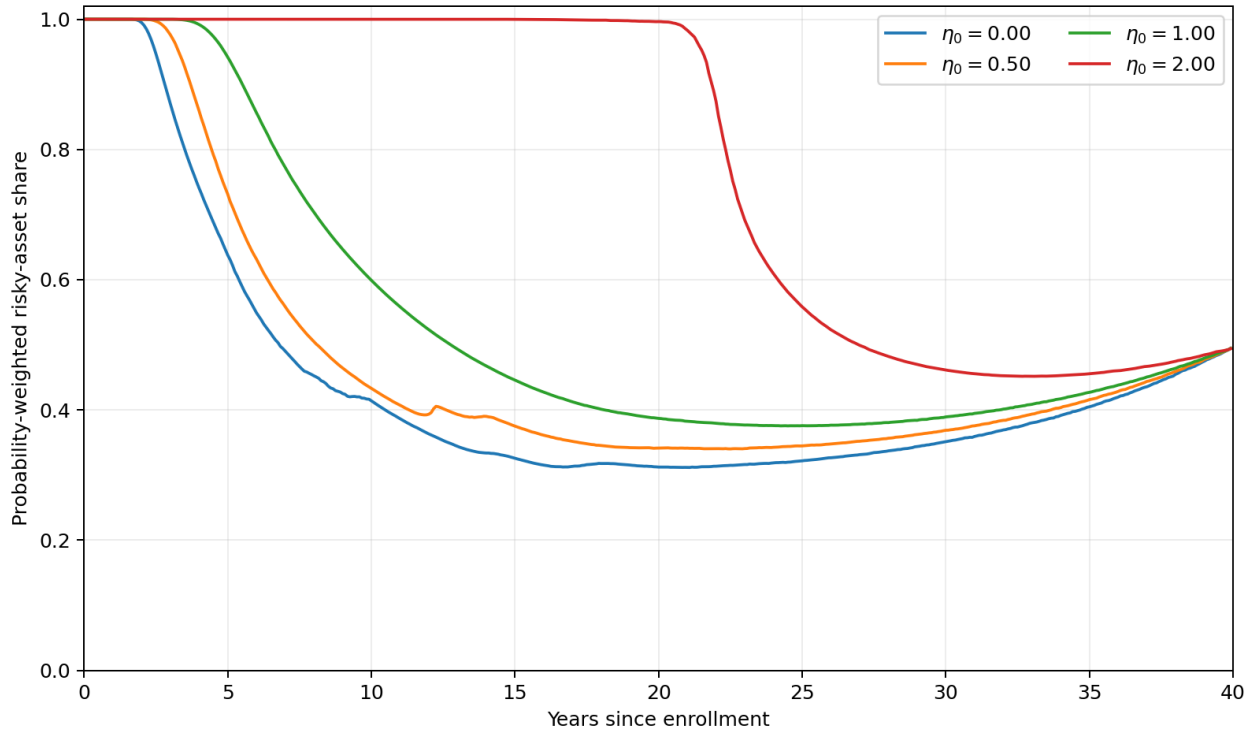


Figure 6: dTCMV-MVS glide paths by skewness coefficient. The variance-aversion coefficient is fixed at $\gamma_0 = 2.5$ and the skewness coefficient takes $\eta_0 = 0, 0.5, 1.0, 2.0$. Each curve is the risky-asset share weighted by the forward wealth distribution generated by that policy.

In Figure 6, $\eta_0 = 0$ corresponds to dTCMV. Increasing the skewness coefficient does not simply raise risky investment uniformly over the whole horizon. At lower positive coefficients, the policy remains at the upper constraint for longer, the middle-horizon trough becomes shallower, and at still higher coefficients a high risky share is maintained for an extended period before converging towards the pre-retirement level. Skewness preference therefore appears as a reallocation of the risk budget across time to create the right tail, rather than as a simple shift towards uniformly more aggressive investment. Because γ_0 is held fixed, the figure isolates the sensitivity of policy shape to η_0 . Mean terminal wealth, however, is not matched across coefficients, so the figure is not a ranking of member outcomes.

The same forward Gauss-Hermite mass propagation can be applied to each of these fixed- γ_0 policies to show the terminal distribution generated by each glide path. A first calculation on the baseline 151-point wealth grid placed all four q05 values on the same grid node, 39.14. That coincidence was numerical rather than economic: the four CDFs crossed the 5% level within the same coarse wealth cell. Because MVS is explicitly treated as mesh-sensitive in this paper, we therefore repeat the fixed- γ_0 experiment on a substantially refined grid, using $N_x = 3601$, $N_a = 97$, and $x_{\max} = 900$, while retaining monthly time steps and five-point Gauss-Hermite quadrature. Table 16 and Figure 7 report the refined outcomes. The density panel is a smoothed visualisation of the discrete terminal probability mass; the CDF and all reported statistics are computed from the unsmoothed probability mass.

Table 16: Refined terminal distribution statistics generated by the fixed- $\gamma_0 = 2.5$ dTCMV-MVS glide paths ($N_x = 3601$, $N_a = 97$, $x_{\max} = 900$)

η_0	Mean	SD	Skewness	q01	q05	Median	q95	LCVaR _{5%}	UCVaR _{5%}
0.00	77.65	22.05	0.865	38.98	47.16	74.72	118.15	42.21	133.31
0.50	79.41	23.69	0.914	38.48	47.03	76.14	123.05	41.84	139.75
1.00	82.82	26.89	1.003	37.75	46.76	78.86	132.67	41.24	152.38
2.00	104.24	60.08	2.294	31.62	41.98	89.21	217.53	35.64	286.49

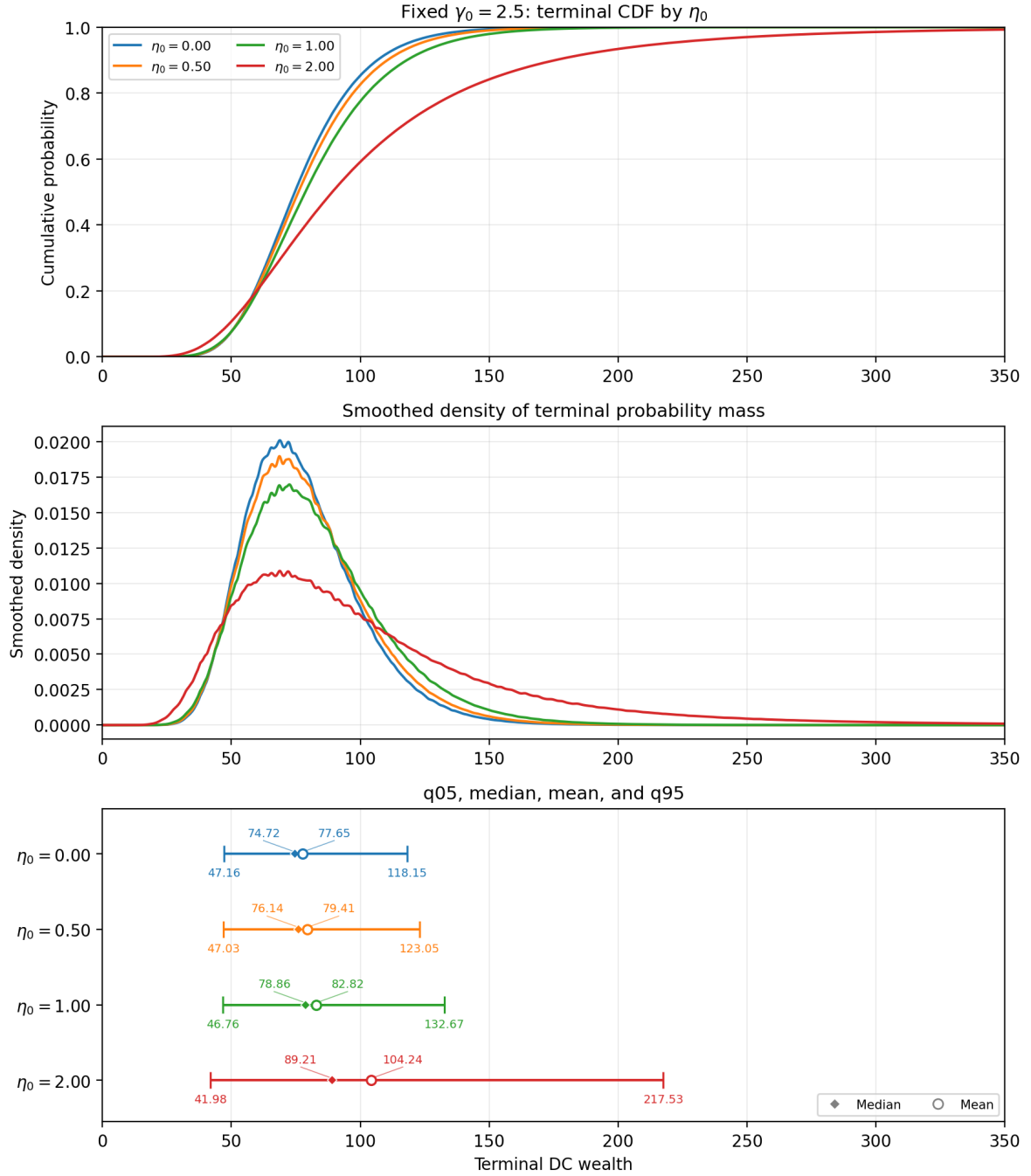


Figure 7: Refined terminal distributions generated by the fixed- $\gamma_0 = 2.5$ dTCMV–MVS glide paths. The upper panel shows the CDFs, the middle panel a smoothed visualisation of terminal probability mass, and the lower panel places the numerical q05, median, mean, and q95 values at their wealth locations. Median and mean labels are aligned at the same height.

The refined calculation confirms that the equal q05 values on the 151-point grid were a discretisation artefact. The q05 values separate to 47.16, 47.03, 46.76, and 41.98 for $\eta_0 = 0, 0.5, 1, 2$, respectively. The refinement also changes the level of several other MVS statistics, which is itself evidence that this non-concave extension remains mesh-sensitive. From $\eta_0 = 0$ to 2, the refined mean rises from 77.65 to 104.24 and q95 from 118.15 to 217.53, while q01 falls from 38.98 to 31.62 and LCVaR_{5%} from 42.21 to 35.64. UCVaR_{5%} rises from 133.31 to 286.49. Thus positive skewness preference strongly expands upside participation but, at high coefficients, also worsens the lower tail. Because this is a fixed- γ_0 coefficient-sensitivity exercise and the results remain discretisation-sensitive, it is not interpreted as a common-mean welfare ranking or as a recommended calibration.

The small oscillations visible in some curves should not be interpreted as economically meaningful short-term trading signals. Once the third central moment is introduced, concavity of the local objective is not guaranteed. The selected control may switch because of small changes in the global search over the control grid, state interpolation, or the propagated distribution. The values $\eta_0 = 0.5, 1.0, 2.0$ are exploratory points, not recommended member calibrations.

PCMV–MVS and DOMV–MVS were also computed but are not used in the operational recommendation. At small skewness coefficients they remain close to their MV counterparts. Beyond a certain coefficient range, how-

ever, especially for PCMV–MVS, numerical branches arise in which the timing of risk-taking changes abruptly. DOMV–MVS is less sensitive overall, but its late-horizon controls remain sensitive to target search, the Lagrange multiplier, control discretisation, and interpolation. These results are retained in the public repository as `supplementary/figures/fig_pcmv_domv_mvs_recalibrated_N80_glidepaths.png`, together with the terminal CDF and coefficient-response figures. Publishing reproducible exploratory results provides a benchmark for future, more robust solvers, but the evidence is not yet sufficient for member-facing implementation.

Table 17: Position of the MVS extensions by solution concept

Extension	What is established here	Practical position
cTCMVS	In the constant-coefficient unconstrained Black–Scholes problem of Mu et al. (2019), it gives the same equilibrium risky amount as cTCMV, and the clipped controls constructed from the common unconstrained solution also coincide. Equality in the directly constrained problem is not established.	No separate operating layer is created for the unconstrained or clipped case. Directly constrained cTCMVS remains a research problem.
dTCMV–MVS	Even with fixed γ_0 , the timing of risk-taking changes materially with η_0 .	Included in the paper as glide-path research, but treated as an exploratory and coefficient-sensitive result.
PCMV–MVS	Close to MV at small coefficients, but numerical branches and abrupt policy changes arise as the coefficient increases.	Retained in the public repository and not used for current plan design or member investment.
DOMV–MVS	Less sensitive than PCMV–MVS, but late-horizon controls remain sensitive to search and discretisation.	Retained in the public repository and not treated as a replacement for the periodic-review layer.

The principal lesson is not merely that a third moment can be added. Higher-moment objectives can alter the timing structure of a glide path nonlinearly and, in some cases, discontinuously. This is important for research but creates substantial governance challenges. MVS should not be introduced into the standard default, periodic-review, or personalised-advice layers until its coefficients can be mapped transparently to retirement-income objectives and preferences, global solutions to the non-concave problem can be independently reproduced, convergence can be assessed by jointly refining time, state, and control grids, and benefits can be confirmed through rolling evaluation and downside-risk constraints. The contribution of this section is therefore to record the cTCMVS equality for the unconstrained and clipped cases while distinguishing the unresolved directly constrained problem; to demonstrate the coefficient-sensitive timing effects in dTCMV–MVS; and to identify the numerical obstacles that future work must overcome for PCMV–MVS and DOMV–MVS.

8 Actuarial Interpretation and Implementation

8.1 From four solution concepts to a three-layer structure

Presenting the four mathematical concepts as four competing products would obscure the practical conclusion. PCMV is not an operating layer: it is a design benchmark for setting terminal targets and organising the mean–variance trade-off. The operating strategies are translated into the three-layer structure shown in Figure 8. MVS is neither a fourth layer nor a present replacement for any existing layer. It remains outside the operating structure as a research extension.

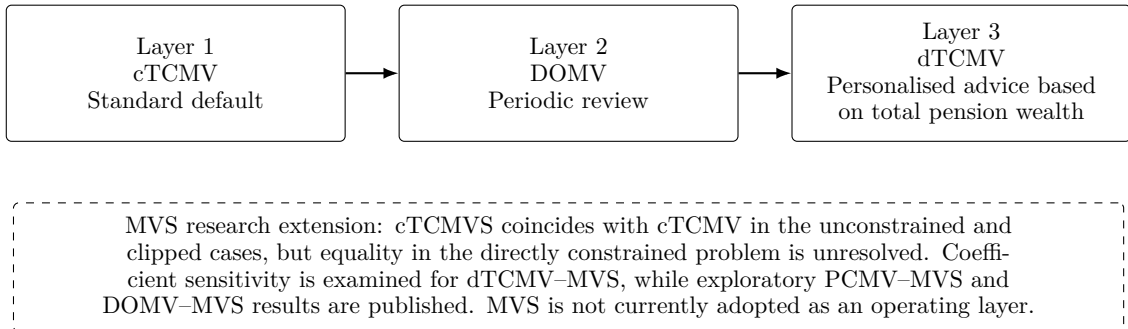


Figure 8: Three operating layers and the position of MVS. MVS is not an additional or replacement operating layer; it is managed outside the three-layer structure until explainable calibration and numerical stability are established.

Implementation principles. cTCMV permits a low-cost clipped implementation, provided that the projection gap is audited periodically. DOMV requires either a directly constrained solver or an updated precomputed policy table at each review date, together with explicit monitoring of the lower tail. dTCMV should be offered as optional personalised advice based on background wealth, account balance, and remaining horizon, with governance of the trade-off between upside participation and additional downside risk. MVS should remain a research extension until coefficient calibration is explainable, non-concave solutions can be independently reproduced, and grid convergence is established.

9 Limitations and Conclusions

9.1 Limitations and future work

The model assumes one risk-free asset and one risky asset, deterministic contributions, constant market coefficients, no transaction costs, and no mortality or unemployment. H_t is the value of future contributions and deterministic benefits; it is not a complete representation of stochastic labour income or human capital. The baseline results are approximations from a monthly discrete-time model on the adopted grids, and comprehensive convergence to a continuous-time solution has not been established.

Positive-coefficient MVS results are exploratory because of non-concavity and grid dependence. For cTCMVS, the established theoretical scope is limited to equality with cTCMV in the unconstrained and corresponding clipped cases; equality in the directly constrained problem is unresolved. PCMV–MVS and DOMV–MVS are therefore preserved as public research outputs rather than operational recommendations. The coefficient-specific dTCMV–MVS glide paths demonstrate sensitivity and are not recommended calibrations. Appendix B establishes weak existence for PCMV and DOMV under stated conditions, but global existence, uniqueness, and regularity of cTCMV and dTCMV equilibria under the two-sided state-dependent constraint remain open.

Appendix A provides external reproduction for the four solution concepts, with dTCMV treated as reconstruction of the published terminal distribution. Backward–forward consistency, mass conservation, and grid diagnostics are available in the repository, but they do not constitute convergence analysis over every possible combination of discretisation parameters. Priority directions are: (i) joint refinement of time, state, control, and Gauss–Hermite order; (ii) cross-validation of DOMV, cTCMV, and dTCMV using independent solvers or alternative discretisation principles; (iii) local refinement of targets, multipliers, controls, and branches in the MVS calculations; and (iv) integration of stochastic income, mortality, multiple assets, costs, and decumulation.

9.2 Conclusions

This paper treats PCMV, DOMV, cTCMV, and dTCMV within the common DC constraint $0 \leq \pi_t \leq X_t$ and distinguishes clearly among unconstrained solutions, clipped approximations, and directly constrained feedback. Although the local problems share a projection structure, the projection in a directly constrained solution is calculated from the constrained continuation value. It is not equivalent to clipping the solution of a different unconstrained problem after the fact.

Using a common Gauss–Hermite-induced transition kernel in a unified backward–forward engine, the paper propagates this distinction through glide paths, terminal distributions, downside tails, and rolling conditional distributions. The law of total variance further decomposes time-zero terminal variance into expected conditional variance remaining within each intermediate state and the cross-state variance of conditional expected wealth. This decomposition motivates a rolling conditional outcome profile that reports conditional means, quantiles, tail risks, skewness, and current investment from an observed intermediate account balance. Own-path and common-state evaluations distinguish remaining-horizon risk, cross-member state heterogeneity, and the prospective effect of the future policy.

Under the baseline calibration, the cTCMV clipping error is small, supporting a low-cost implementation for the standard default. DOMV’s periodic re-optimisation is not reproduced reliably by clipping, so a directly constrained solver is required. dTCMV adjusts upside participation according to total pension wealth but trades it against downside risk; it should therefore be governed as optional personalised advice. PCMV remains the design benchmark.

The practical conclusion is to place cTCMV, DOMV, and dTCMV in three operating layers while retaining PCMV as the design standard. MVS is neither a fourth layer nor a replacement for an existing layer. Given the cTCMVS equality in the unconstrained and clipped cases but unresolved directly constrained problem, the large coefficient sensitivity of dTCMV–MVS, and the numerical instability of PCMV–MVS and DOMV–MVS, it remains outside the three-layer structure as a research extension. The value of the framework is not a universal preference for complexity. It is the ability to identify when inexpensive clipping is adequate and when rigorous constrained optimisation creates additional value for member outcomes and governance.

A Code, Numerical Data, and Validation Materials Supporting Reproducibility

A.1 Public repository and reference version

The Python code, principal CSV and NPZ data, figure-generation scripts, and execution instructions are published in the English-language repository:

<https://github.com/yoshi123027-del/dc-glidepath-ica2026-en>

To identify the implementation corresponding to this English manuscript even after later repository updates, the documentation reference is fixed at:

<https://github.com/yoshi123027-del/dc-glidepath-ica2026-en/commit/a3b83a590474a4d88642d7ce95cfddda9b60461a>

The public materials cover the baseline case, technical appendices, the PCMV and DOMV solvers, extended-HJB calculations for cTCMV and dTCMV, MVS extensions, rolling conditional evaluation, comparison between directly constrained feedback and clipped approximations, the U-shaped decomposition in Table 8, and the code and numerical data required to regenerate the figures. No personal data, confidential data, or external authentication credentials are included.

Figures supplementing the main analysis are collected at:

<https://github.com/yoshi123027-del/dc-glidepath-ica2026-en/tree/a3b83a590474a4d88642d7ce95cfddda9b60461a/supplementary/figures>

The supplementary collection includes terminal-wealth densities, upper-constraint binding probabilities and free boundaries, projection gaps, rolling conditional evaluations, sensitivity analysis for principal parameters, and detailed MVS results. Each figure is accompanied by an English explanation of the numerical fact, its mathematical and practical implications, and interpretive cautions. These materials do not replace the conclusions of the paper; they support robustness assessment and numerical transparency.

A.2 Structure of the public materials

The repository organises solvers, calibration, rolling evaluation, sensitivity analysis, figure generation, and validation in functional directories. It supports both complete recomputation and rapid figure regeneration from stored CSV and NPZ files. Dependencies, execution order, and outputs are documented in `README.md`, `scripts/README.md`, and `CODEBOOK.md`; detailed validation and diagnostic instructions are provided in `validation/README.md` and `diagnostics/README.md`. The manuscript provides only the entry point, while the repository documentation is the source of record for the detailed file inventory and commands.

A.3 External validation for the four MV solution concepts

This appendix reproduces external numerical examples for PCMV, DOMV, cTCMV, and dTCMV. Details of backward-forward consistency, mass preservation, grid stability, and automated regression tests for the directly constrained implementation are contained in the public repository. Table 5.1 of [van Staden et al. \(2021\)](#) reports terminal-wealth statistics for all four concepts under a common market setting and two target means, so the external reproduction is not restricted to PCMV.

The amount of published information differs by concept. For PCMV, DOMV, and cTCMV, the distributions can be recomputed independently from the published closed forms. For dTCMV, the paper establishes a lognormal terminal distribution but does not report the numerical path of the time-varying coefficient. The validation therefore combines reconstruction of the published dTCMV terminal distribution with internal diagnostics of the constrained solver.

A.3.1 External reproduction of Table 5.1 in van Staden et al. (2021)

Following Section 5 of that paper, set initial wealth $w_0 = 100$, horizon $T = 10$ years, $\mu = 0.0816$, $\sigma = 0.1863$, and $r = 0.00623$, with no contributions, transaction costs, or leverage constraint. This is an audit of the common unconstrained special case, not a recalibration of the 40-year contributed baseline under $0 \leq \pi \leq X$. The target means are 125 and 250.

For PCMV, the reflected-lognormal distribution is recomputed from equations (3.6) and (4.2) of the reference paper. For DOMV and cTCMV, the normal terminal distribution in equations (3.12)–(3.15), together with the variance-aversion coefficients associated with the target means in equations (4.3)–(4.4), gives the mean, variance, and tail measures in closed form. For dTCMV, equation (3.16) gives a lognormal terminal distribution. Because the numerical solution of the integral equation determining its time-varying coefficient is not reported in Table 5.1, the published mean $\bar{w}$ and standard deviation s_w identify the lognormal distribution through

$$s_L^2 = \log\left(1 + \frac{s_w^2}{\bar{w}^2}\right), \quad \mu_L = \log \bar{w} - \frac{1}{2}s_L^2. \quad (117)$$

The median, skewness, excess kurtosis, VaR, lower-tail CVaR, shortfall probabilities, and conditional means are then recomputed independently. This is a reconstruction of the published terminal distribution, not an independent reproduction of the unpublished time-varying coefficient path. That control path is assessed separately through the backward–forward and grid diagnostics in the repository.

Table 18: Reproduction of Table 5.1 in van Staden et al. (2021): published/reproduced values

Target mean	Model	Parameter	Mean	Median	Standard deviation	VaR _{5%}	LCVaR _{5%}
125	PCMV	259/258.976	125/125.000	127/127.508	9/9.130	113/113.250	97/97.412
	DOMV	0.111/0.111411	125/125.000	125/125.000	16/15.994	99/98.692	92/92.009
	cTCMV	0.044/0.044064	125/125.000	125/125.000	15/14.517	101/101.122	95/95.056
	dTCMV	0.041/–	125/125.000 ^a	124/123.988	16/16.000 ^a	101/100.535	96/95.424
250	PCMV	569/569.388	250/250.000	269/269.389	71/70.577	159/159.165	37/36.726
	DOMV	0.014/0.014412	250/250.000	250/250.000	124/123.643	47/46.626	–5/–5.039
	cTCMV	0.006/0.005700	250/250.000	250/250.000	112/112.223	65/65.409	19/18.515
	dTCMV	0.001/–	250/250.000 ^a	123/122.658	444/444.000 ^a	17/17.227	11/11.342

^a The dTCMV mean and standard deviation are anchors used to identify the published terminal distribution through (117); their agreement is therefore not an independent reproduction result.

Because Table 5.1 is reported mainly to integer precision, the tolerance is one unit for published and reproduced non-probability statistics and 0.011 for probabilities. The tolerance for decimal variance-aversion parameters is 6×10^{-4} . In the Monte Carlo check, the default tolerances are 5×10^{-4} for probabilities, $\max(0.05, 0.004|q|)$ for means, medians, and standard deviations, $\max(0.75, 0.01|q|)$ for the most sampling-sensitive one-to-three-percent lower-tail quantities, and $\max(0.35, 0.006|q|)$ for other tail statistics, with wider allowances for the thick right tail of dTCMV; q denotes the reproduced value.

The representative indicators in Table 18 agree, for PCMV, DOMV, and cTCMV, with recalculation from the published closed forms and with the corresponding Monte Carlo results within the stated tolerances. For dTCMV, the median, quantiles, and tail statistics are reproduced from the lognormal distribution identified by the published mean and standard deviation. This confirms consistency of the published terminal distribution, not independent reproduction of the unpublished time-varying coefficient path. Full diagnostic values, tolerances, and internal-audit outputs are stored in the repository CSV files.

B Technical Supplement on Weak Existence and Verification for PCMV and DOMV

This appendix first separates weak existence for PCMV and DOMV from any assumption of a classical solution. It then states explicitly the verification argument used for PCMV and DOMV in Section 4, and concludes by collecting the free-boundary and projection structure common to the four concepts. Weak existence relies on standard theory for controlled diffusions, dynamic programming, and viscosity solutions, whereas the local feedback representation and classical verification require candidate functions of class $C^{1,2}$ and suitable integrability. No claim is made here that the free boundaries specific to this paper possess a unique classical solution or are smooth.

B.1 Weak existence for PCMV and DOMV

Assume that $c : [0, T] \rightarrow [0, \infty)$ is bounded and continuous and that $D_T \geq 0$ is constant. For $x > 0$, set $u_s = \pi_s/X_s$, while for $x = 0$ set $u_s = 0$. Let u be a progressively measurable process with values in $[0, 1]$. The state equation becomes

$$dX_s = \{c_s + (r + \beta u_s)X_s\} ds + \sigma u_s X_s dW_s, \quad X_t = x \geq 0. \quad (118)$$

For a fixed target m , define

$$L^m(t, x) = \inf_u \mathbb{E}_{t,x}[(X_T^u + D_T - m)^2]. \quad (119)$$

Proposition 3 (Weak existence for PCMV and DOMV). *Under the conditions above:*

- (i) *For every normalised control u , (118) has a unique non-negative strong solution. For every $p \geq 2$ and every compact set of initial states, $\mathbb{E}[\sup_{t \leq s \leq T} |X_s^u|^p]$ is uniformly bounded over controls.*
- (ii) *L^m is finite, continuous in (t, x, m) , and of at most quadratic growth. It satisfies the dynamic programming principle and is a continuous viscosity solution of HJB (20). If comparison holds in the class of continuous viscosity solutions with polynomial growth, L^m is unique in that class. An optimal Markov control exists in the compactified class of relaxed controls. If a Borel-measurable strict selector of the HJB minimiser exists and the resulting closed-loop SDE is well posed, an optimal strict Markov control can be selected.*
- (iii) *For $\gamma > 0$, define*

$$\Psi_\gamma(t, x; m) = m - \frac{1}{2\gamma} - \frac{\gamma}{2} L^m(t, x). \quad (120)$$

Then $m \mapsto \Psi_\gamma(t, x; m)$ is continuous and tends to $-\infty$ as $|m| \rightarrow \infty$. The outer maximisation in scalarised PCMV therefore has at least one target $m_\gamma^(t, x)$.*

(iv) The remaining-horizon DOMV problem has a non-empty compact set of optimal targets

$$\mathcal{M}_D^*(t, x) = \arg \max_{m \in \mathbb{R}} \Psi_{\gamma_D}(t, x; m).$$

A Borel-measurable selector $m_D^*(t, x)$ can be taken from this correspondence. By connecting the current component of an optimal Markov control for each selected target, a DOMV feedback can be constructed in the weak sense.

The proposition does not assert:

- (a) that an embedding target satisfying mean consistency exists for every arbitrarily prescribed target mean d ;
- (b) that the value function is of class $C^{1,2}$;
- (c) that the free boundary is smooth; or
- (d) that an optimal strict feedback is unique and continuous.

Proof. Normalisation makes the control set the fixed compact interval $[0, 1]$. The drift and diffusion coefficients in (118) are uniformly Lipschitz in x and have linear growth. A unique strong solution therefore exists for every progressively measurable control. With the stochastic exponential

$$Z_s = \exp \left(\int_t^s \left(r + \beta u_q - \frac{1}{2} \sigma^2 u_q^2 \right) dq + \int_t^s \sigma u_q dW_q \right),$$

we have

$$X_s = Z_s \left(x + \int_t^s Z_q^{-1} c_q dq \right) \geq 0. \quad (121)$$

The Burkholder–Davis–Gundy and Gronwall inequalities yield the uniform moment bound in (i).

The terminal loss is continuous and has quadratic growth, and the control set is compact. Standard finite-horizon controlled-diffusion theory therefore gives finiteness and continuity of the value function, the dynamic programming principle, and the viscosity-solution property (Fleming and Soner, 2006). Compactifying a sequence of strict controls in the relaxed-control class yields an optimal relaxed Markov control. When a Borel selector of the HJB minimiser exists and the selected closed-loop equation is well posed, the standard verification and selection argument recovers a strict Markov control. This proves (ii).

The moment estimate gives a control-uniform bound on the first moment of reachable terminal wealth $W_T^u = X_T^u + D_T$. For some $C_{t,x} < \infty$, $\sup_u \mathbb{E}_{t,x}[W_T^u] \leq C_{t,x}$. Jensen's inequality therefore gives

$$L^m(t, x) \geq \inf_u \{ \mathbb{E}_{t,x}[W_T^u] - m \}^2 \geq (|m| - C_{t,x})_+^2.$$

Thus (120) tends to $-\infty$ as $|m| \rightarrow \infty$. Together with continuity of L^m , this proves existence of a maximiser and hence (iii). The same estimate applies to every remaining-horizon problem. Joint continuity and locally uniform coercivity of Ψ_{γ_D} make the maximiser correspondence non-empty, compact-valued, and measurable. The measurable maximum theorem gives a Borel selector m_D^* , and connecting the current controls of the selected target problems proves (iv). $\square$

The purpose of this proposition is to separate existence of PCMV and DOMV from existence of a classical solution with a smooth free boundary. Li et al. (2002) and Li and Xu (2015) provide the viscosity-solution and optimal-strategy background for constrained mean–variance problems, while Guan et al. (2022) establish a related classical solution and regular free boundary under a borrowing constraint. The latter is a CRRA utility problem and does not transfer classical existence directly to the quadratic-loss HJB used here.

B.2 Classical verification for PCMV and DOMV

For a fixed target m and any admissible control π , let τ_k be the minimum of a bounded-domain exit time and a localisation time controlling quadratic variation. Itô's formula gives

$$\mathbb{E}[v^m(\tau_k, X_{\tau_k}^\pi)] - v^m(t, x) = \mathbb{E} \int_t^{\tau_k} (v_s^m + \mathcal{L}^{\pi_s} v^m)(s, X_s^\pi) ds. \quad (122)$$

By HJB (20), the integrand is non-negative for every π and is zero for a minimising control $\pi^{m,*}$. Passing to the limit $k \rightarrow \infty$ using polynomial growth and uniform integrability, and applying the terminal condition, yields

$$v^m(t, x) \leq \mathbb{E}_{t,x}[(X_T^\pi + D_T - m)^2], \quad v^m(t, x) = \mathbb{E}_{t,x}[(X_T^{\pi^{m,*}} + D_T - m)^2].$$

The part of the Hamiltonian depending on π is

$$q^m(\pi) = \beta v_x^m \pi + \frac{1}{2} \sigma^2 v_{xx}^m \pi^2.$$

Under $v_{xx}^m > 0$, this is convex, so the constrained minimiser is (21). Mean-constrained PCMV reduces to this quadratic-loss minimisation through the identity stated in the main text. DOMV applies the same argument to the remaining-horizon problem at every (t, x) and connects the first control diagonally.

Free boundaries and the unified projection structure

For PCMV, cTCMV, and dTCMV, each local Hamiltonian is a one-dimensional convex or concave quadratic under the relevant effective-curvature condition. The state space separates into a lower-bound region, an interior region, and an upper-bound region according to whether the unconstrained stationary point lies below 0, inside $[0, x]$, or above x . Because the stationary point is computed from derivatives of the constrained value function or equilibrium moments, the switching set is a free boundary depending on unknown functions. This common structure unifies the projection form but does not make a clipped unconstrained solution identical to the directly constrained feedback.

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